

# **Data visualization** guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 0.2 (pilot) September 2026 All example data is synthetic

OVERVIEW

# How to use this guide

One set of rules, three ways in. The guide teaches the thinking; the starter kit makes the right answer the easy one in every tool we use.

Every chart we make should do one job: answer a business question so clearly that the reader knows what to do next. The rules here are written to be checked, by a reviewer or by an agent, before a chart leaves the team.

Inspired by the structure of a leading consulting firm's public data-visualization guide, rebuilt for McKesson's brand and for the work of Workforce Innovation.

### CONSULTANTS AND ANALYSTS

**Decks, reports, workbooks**

Read Fundamentals once, then use the chart pages as you build. Keep the cheat sheet next to you.

- PowerPoint template and starter deck
- Excel chart workbook with the palette
- One-page cheat sheet and chart chooser
- Power BI report theme

### DEVELOPERS AND DATA SCIENTISTS

**Python, HTML, R, Power BI**

Each chart page links to tested code. Themes are generated from one token file, so every tool matches.

- `mckviz` for matplotlib and Plotly
- CSS variables, Plotly and Vega-Lite themes
- `theme_mck()` for ggplot2
- `mv.lint(fig)` to check a chart

### CODING AGENTS

**Claude, Copilot and others**

Agents read `AGENTS.md` automatically in this repo. For other repos, copy one self-contained rules file.

- `AGENTS.md`: workflow and rules
- `mck-chart` skill: question to QA
- `dist/agents/mck-viz-rules.md`
- Cookbook pages with runnable snippets

### The five steps behind every chart

**1 Question.** Write the question the chart answers in one sentence.

**2 Chart.** Pick the chart from the question type, not from habit.

**3 Build.** Grey first, then color only what the headline names.

**4 Headline.** Compute the finding, then write it as a sentence.

**5 QA.** Run the checklist and look at the rendered chart.

OVERVIEW

# Contents

Fundamentals apply to every chart. Each chart family then gets the same treatment: anatomy, variants, best practice and how to build it.

This is the pilot release. Sections marked NEXT follow the same pattern once the look and depth of the pilot are agreed.


01 **Fundamentals**
Question first, how readers read, anatomy, headlines, color, type, clutter, honest scales, people data, accessibility

02 **Magnitude**
Bar and column charts: anatomy, variants, best practice, build it

03 **Part-to-whole**
Stacked bars, waterfalls, treemaps, donuts: anatomy, variants, best practice, build it

04 **Change over time**
Line, stacked area, bubble timeline, small multiples

05 Ranking NEXT
Slope, bump, dot plot and dumbbell, heat table

06 Distribution NEXT
Histogram, strip, box plot

07 **Correlation**
Scatter and bubble, heatmaps, why not dual axes

08 **Relationships and flows**
Sankey, parallel coordinates, Venn

09 Geo-spatial NEXT
Choropleth, bubble map, dot density

10 Single number NEXT
KPI and big number, progress to target

11 Formats NEXT
Slides, reports and print, web and mobile, dashboards, interactivity, animation, social

12 Toolkit NEXT
Downloads, set-up for each tool, the QA checklist, guidance for agents

# **Start with the question,** not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

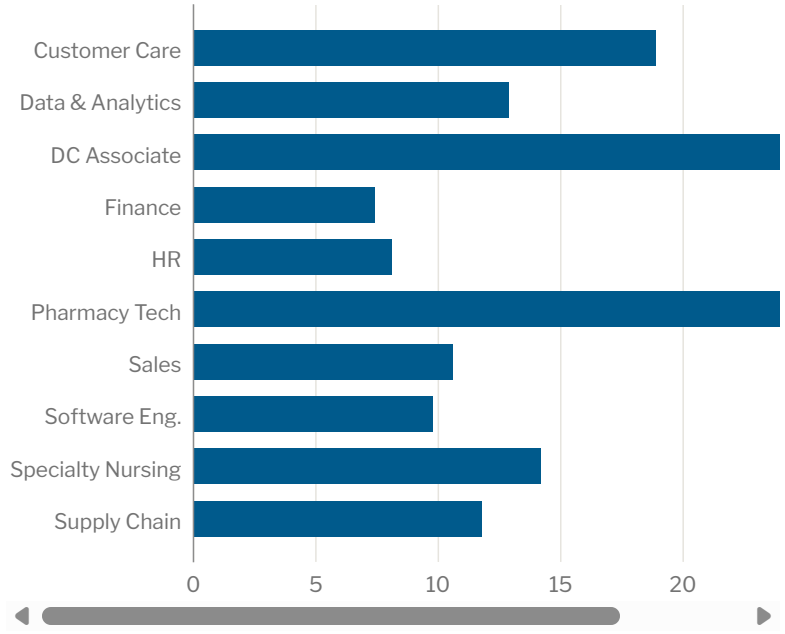
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

### Attrition by job family

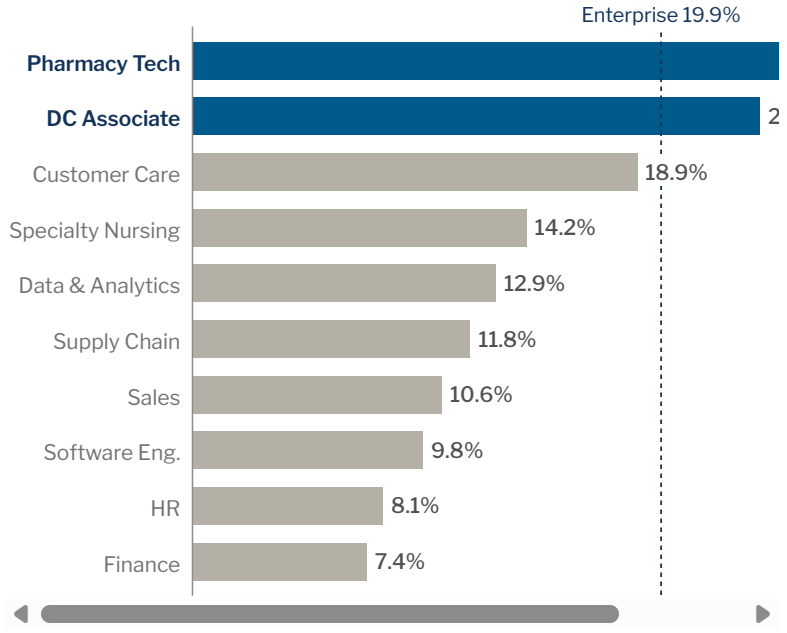
FY26




**Shown**
A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

### Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)




**Explained**
The headline states the finding, the order ranks the families, and color points to the two the headline names.

# **How readers** read a chart

Readers move through a chart in three steps. Design for each one, in order.

### 1 Perceive

In the first second the eye takes in shape, length and color before reading a word. Make the thing that matters the thing that stands out: one color against grey.

### 2 Interpret

Next the reader looks for meaning: headline, labels, units, benchmarks. The headline should confirm what they just saw, or sharpen it.

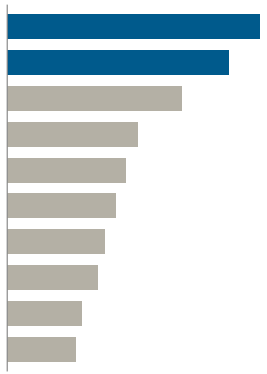
### 3 Act

Finally they ask "so what?". A callout that sizes the implication turns a finding into a decision.

STEP 1 · PERCEIVE

**Two bars stand out**

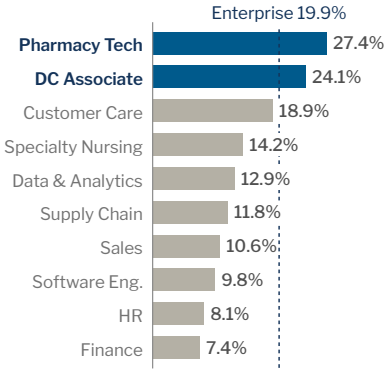
No words yet. Grey context and blue focus already tell the reader where to look.



STEP 2 · INTERPRET

**Labels give it meaning**

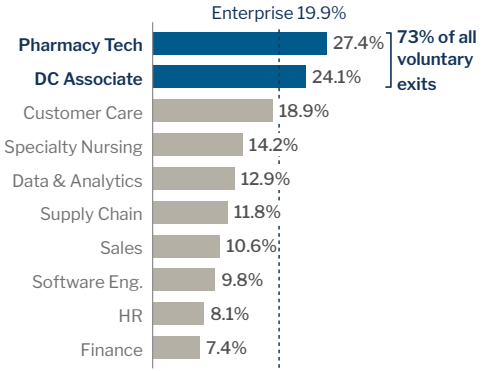
Names, values and a labelled benchmark show the two families sit well above 19.9%.




STEP 3 · ACT

**The callout sizes the prize**

Together they hold 58% of headcount but 73% of voluntary exits: a retention plan for two families moves the enterprise number.



# **Anatomy** of a chart

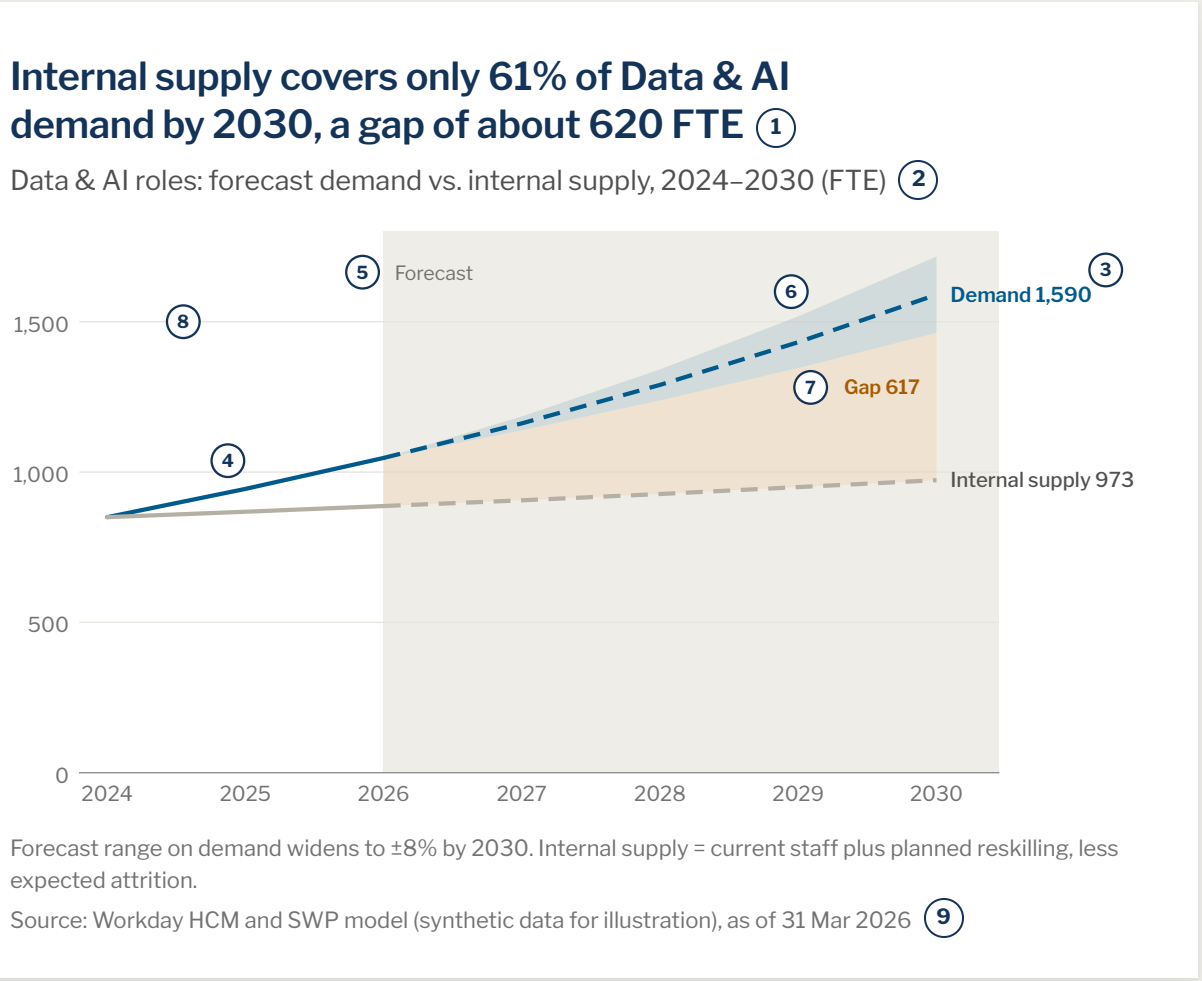
Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.



FUNDAMENTALS

# **Write the headline** last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

### The formula

**Who or what + verb + number + so what.**

### Before you keep it

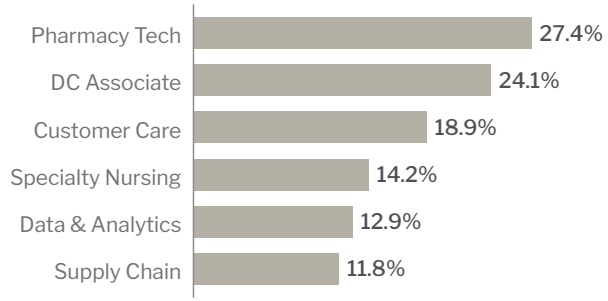
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

| | |
|---|---|
| **LABEL** | Attrition by job family · names the topic; says nothing |
| **VAGUE** | Attrition varies across job families · true of every chart ever made |
| **OPINION** | Attrition is out of control in the frontline · a claim the chart can't support |
| **INSIGHT** | **Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate** |
| **INSIGHT + ACTION** | **Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate** |

## Attrition varies across job families

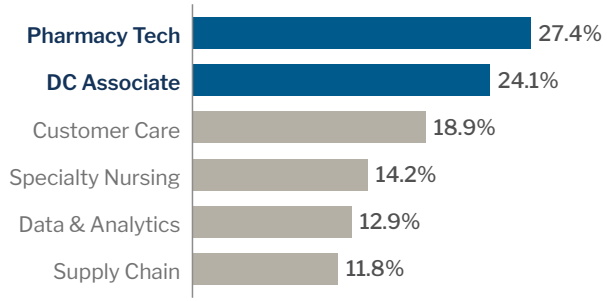
Voluntary attrition by job family, FY26 (%)




**Headline and chart both neutral**
Nothing tells the reader where to look.

## Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)




**Headline and highlight agree**
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

# **Grey first,** then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

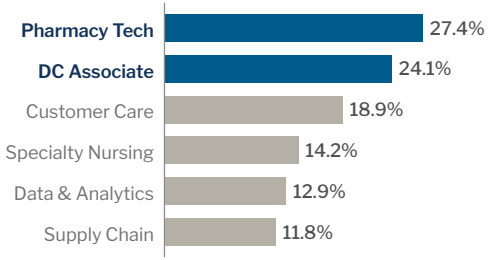
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

**Orange means "worse".** Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

### Focus

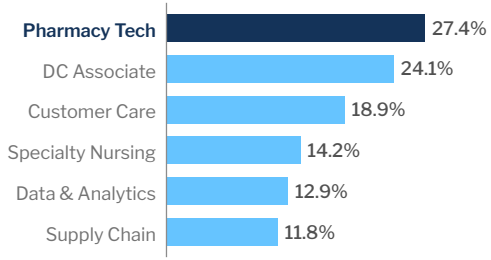
All grey; the items the headline names in blue.




PATTERN 2

### Uniform + accent

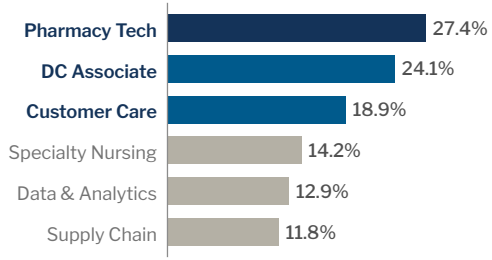
One series in light blue; the single exception in navy.




PATTERN 3

### Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.




### Color roles

| | | |
|---|---|---|
| **Context** · everything the reader should see but not focus on | `#B4B0A5` |
| **Focus** · what the headline names | `#005A8C` |
| **Accent** · the one exception that must pop (navy, not orange) | `#143359` |
| **Light blue** · the series when navy marks the exception | `#66C4FF` |
| **Orange** · only when it means worse: below target, shrinking, a shortfall | `#EF8200` |
| **Total and benchmark** · totals, targets, reference lines | `#143359` |
| **Context light** · de-emphasized fills in stacks and areas | `#DEDBD1` |

# **Three palettes,** three jobs

Pick the palette from what the color encodes.

### Categories

Only when categories must be told apart. At most six, always in the same order, so "Full-time" is the same blue on every slide.

### One measure, light to dark

For heat tables and choropleths of a single measure. Darker means more.

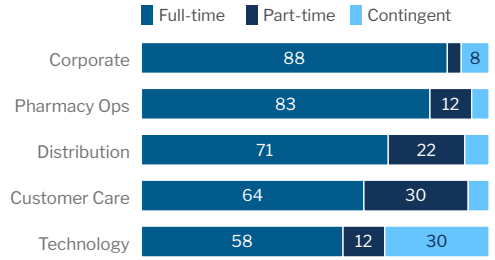
### Gap to a benchmark

For values centered on zero or a target. Orange below, blue above, a neutral middle.

Text on a fill switches between white and navy, whichever contrasts more (`mv.text_on`).

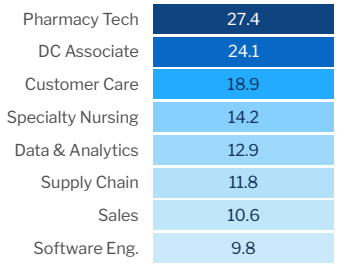
**CATEGORICAL**

Workforce mix by function (%)



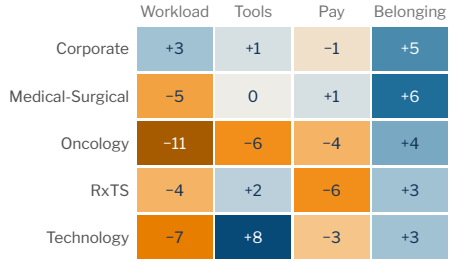

**SEQUENTIAL**

Voluntary attrition, FY26 (%)




**DIVERGING**

Engagement vs. healthcare benchmark (pts)




**Categorical**
Max 6, always this order

005A8C 143359 66C4FF 00AF9D 00737A EF8200

**Sequential**
One measure, light to dark

E6F3FB B6E2FA 66C4FF 009EFF 0A5FBC 143359

**Diverging**
Centered on 0 or a target

A75B00 EF8200 F7C180 EFEDE7 99BDD1 005A8C 143359

# **Type** and proportion

One typeface, Libre Franklin, in a few fixed roles. Type should frame the data, not compete with it.

The headline is the largest text on the chart but the data is the largest *thing*. If the headline is the first thing you notice from across the room, it is too heavy.

- Nothing below 9 pt on slides, 11 px on screens
- Sentence case everywhere; no all-caps headlines
- Headline, subtitle, key, footnote and source share one left edge
- Big numbers use the light weight

Libre Franklin is open source (SIL OFL). Set-up for Office, the web and Power BI is in the Toolkit.

| Role | Sample | Weight | Slide | Screen |
|---|---|---|---|---|
| Headline | **Attrition is concentrated in two job families** | Bold | 24 pt | 22 px |
| Subtitle | Voluntary attrition, FY26 (%) | Regular | 14 pt | 15 px |
| Data labels | 27.4% | Medium | 12 pt | 13 px |
| Axis and category labels | Pharmacy Technician | Regular | 11 pt | 12 px |
| Callouts | Hiring freeze, Q2 | Regular | 12 pt | 13 px |
| Footnote and source | Source: Workday HCM, as of 31 Mar 2026 | Light | 9 pt | 11 px |
| Big number | 75% | Light | 54 pt | 56 px |

### Proportion on a 16:9 slide

Margins of about 0.5 in. The chart takes at least 60% of the slide height. One message per chart, at most two charts per slide; the slide title is the headline.

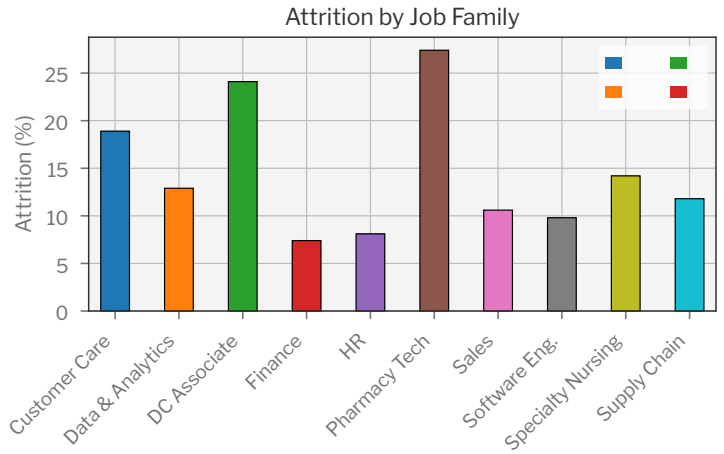
### Proportion on a screen

The chart card sits on a quiet background with generous padding. Headline about 1.4× the subtitle. On phones, swap end labels for a top key and let wide exhibits scroll sideways rather than shrink.

# **Label directly,** delete the rest

Every element must earn its place. If removing it loses nothing, remove it.

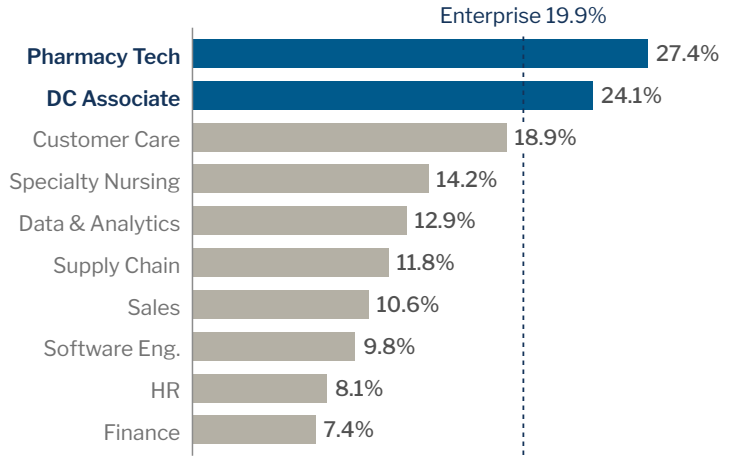
1. Replace the label-style title with an insight headline
2. Sort by value, not alphabet
3. Turn long-named columns into horizontal bars
4. Grey everything, then color the point
5. Label values in place; delete the value axis, gridlines and tick marks
6. Delete the legend, borders and background fill
7. Add the benchmark, labelled where it sits




**Tool defaults**
Rainbow fills, borders, rotated labels, gridlines, a legend that decodes nothing.

**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**

Voluntary attrition by job family, FY26 (%)




**Decluttered**
The same data. Fewer elements, and every one of them carries information.

FUNDAMENTALS

# **Honest** scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

**Bars start at zero.** If the differences only show on a narrow range, change the chart, not the axis.

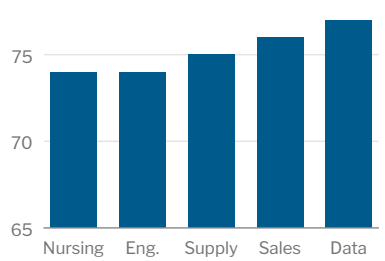
**No dual axes.** Two scales let you make any two lines cross wherever you like.

**Bubble area, not diameter,** is proportional to value.

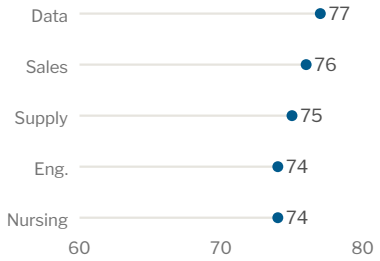
**Pies take five slices at most,** and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

## Truncated bars

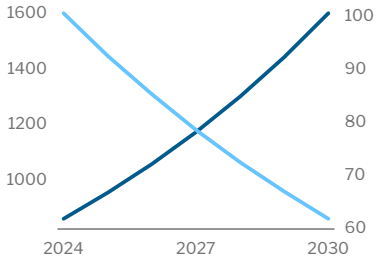



**Try to avoid**
An axis from 65 turns a 4% difference into a 33% one.

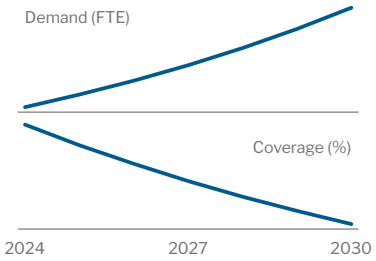



**Recommended**
A dot plot compares position, so the range can be narrow.

## Dual axes

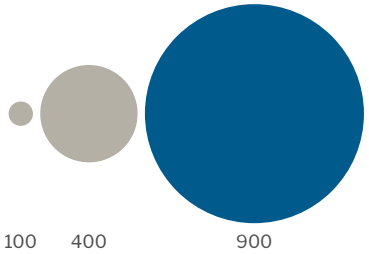



**Try to avoid**
The crossing point is an artifact of two chosen scales.

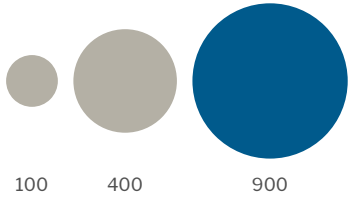



**Recommended**
Two aligned charts share the time axis.

## Bubble size

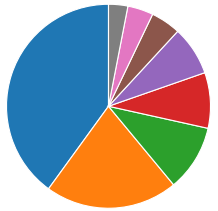



**Try to avoid**
Sized by diameter, 900 looks 81× bigger than 100.

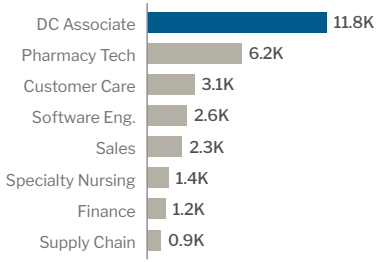



**Recommended**
Sized by area, 900 is 9× bigger, as it should be.

## Too many slices



**Try to avoid**
Eight similar angles can't be ranked by eye.




**Recommended**
A sorted bar ranks them instantly.

# **People data** is personal

Workforce charts describe colleagues, and carry extra obligations.

### Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

### No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

### Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

**Engagement by team, one distribution center**

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Returns | 7 | 52 |
| Quality | 4 | 88 |
| Site leadership | 9 | 81 |

**Try to avoid**
Reporting groups below the minimum, even in a table or tooltip.

**Engagement by team, one distribution center**

Small teams combined into one group that clears the threshold.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Other teams (3 combined) | 20 | 72 |

**Recommended**
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

### In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

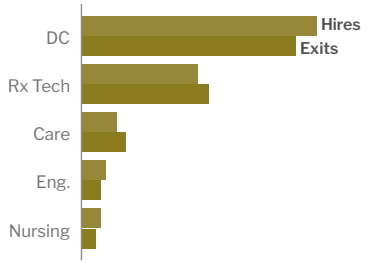
# **Accessible** by default

About 1 in 12 men and 1 in 200 women have a color-vision deficiency. In a room of 25, assume someone does.

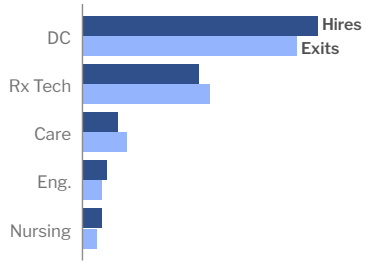
- Text contrast at least 4.5:1
- Never red vs. green; use blues and navy, with orange only for "worse"
- Never color alone: pair it with labels, position or shape
- Nothing below 9 pt on slides or 11 px on screens
- Alt text for every chart that leaves the team: the headline plus two or three key numbers

The red–green examples are simulated for deuteranopia, the most common form.

## Red vs. green

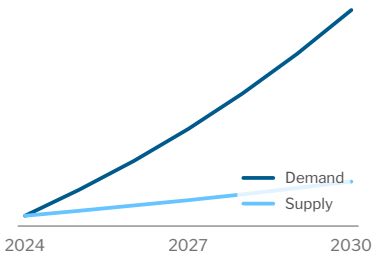



**As seen with deuteranopia**
Red and green hires vs. exits become one color.

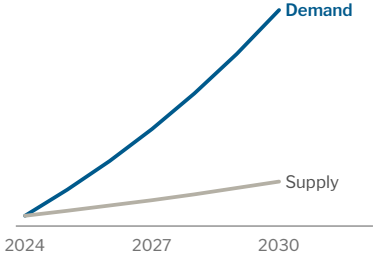



**As seen with deuteranopia**
Blue and light blue stay distinct.

## Color as the only cue

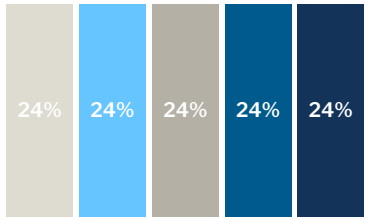



**Try to avoid**
The reader matches colors to a legend.

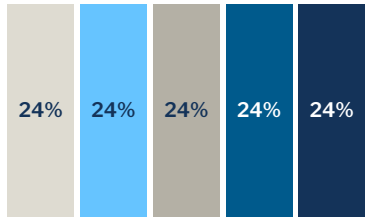



**Recommended**
Names at the line ends; color only reinforces.

## Text on fills




**Try to avoid**
White on every fill fails on the light ones.




**Recommended**
White or navy, whichever contrasts more.

## Alt text

Write it from the headline and the numbers that prove it.

> **Alt:** Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate. Pharmacy Technician 27.4%; Distribution Center Associate 24.1%; all other job families are below the enterprise rate.

In PowerPoint: right-click the chart, View Alt Text. In HTML: `aria-label` on the figure.

MAGNITUDE

# **Bar** chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

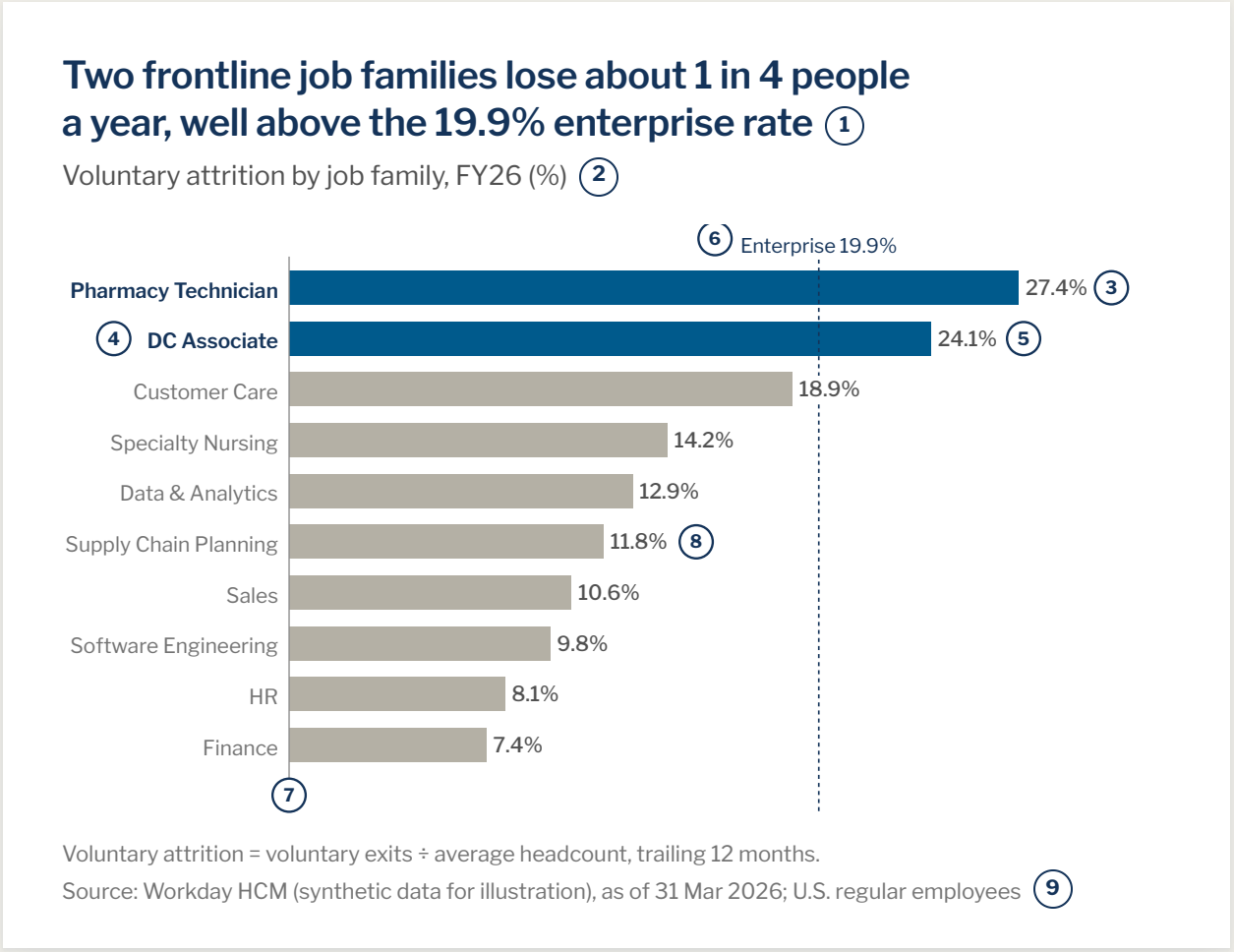
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

### McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

# Bar chart variants

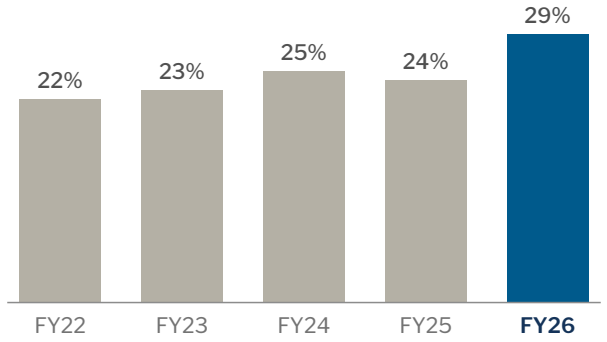
Same rules, different shapes. Pick the variant from the comparison you need to make.

### When a bar is the wrong chart

- Values bunched far from zero (engagement 71–78): use a **dot plot**
- Largest more than 20× the smallest: use **proportional bubbles** or split the chart
- Parts of a whole: use a **100% stacked bar** or **waterfall**
- Many periods: use a **line**
- Spread across people: use a **distribution**

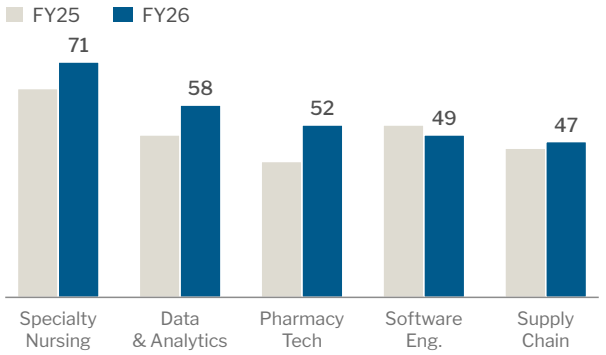
### Column chart

A few periods in time order, or short category names. Here internal fill rose from 22% to 29%.



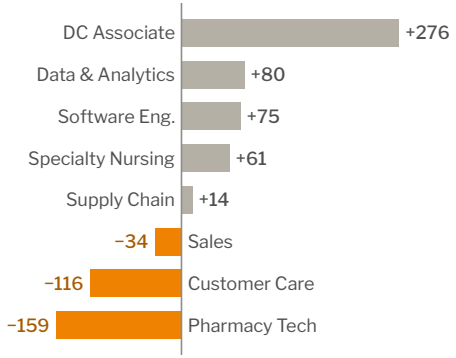

### Grouped bars

Two or three series per category. Prior period in light grey, current in blue; key top-left.



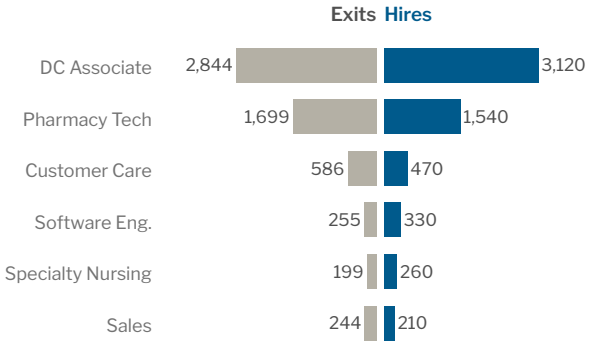

### Diverging bars

Values on both sides of zero. Three job families shrank in FY26 despite hiring; negatives in orange.




### Butterfly chart

Two measures back to back on the same items: FY26 exits and hires by job family.



MAGNITUDE

# **Bar chart** best practice

Four rules that fix most bar charts.

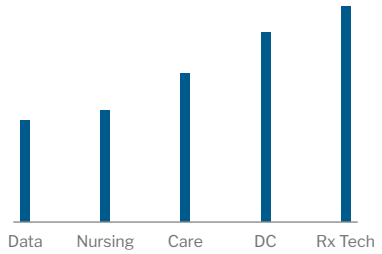
**Size of bars.** Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

**Color of bars.** A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

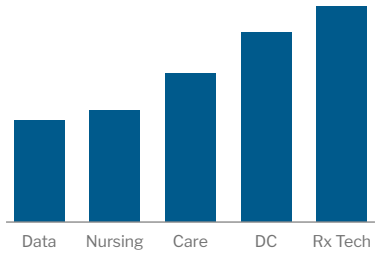
**Direct labels.** Gridlines and a value axis add noise. Label the bars and delete both.

**Zero baseline.** Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

## Size of bars

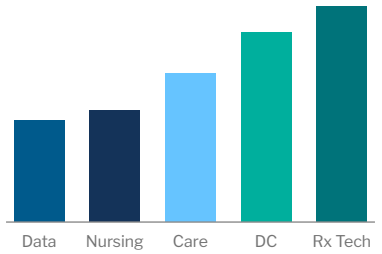



**Try to avoid**
Stick-thin bars; the gaps dominate.

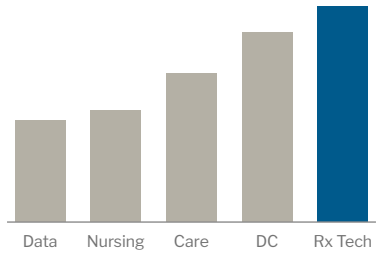



**Recommended**
Bar about twice the gap.

## Color of bars

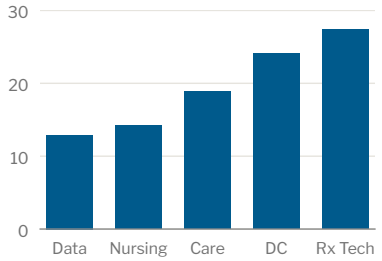



**Try to avoid**
A color per bar, encoding nothing.

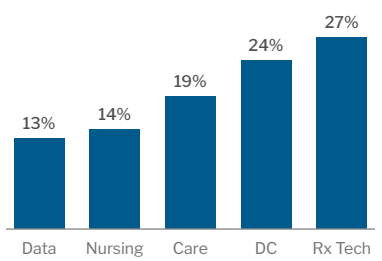



**Recommended**
Grey bars; blue for the one that matters.

## Direct labels

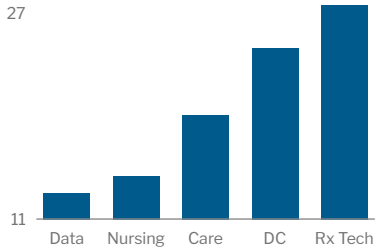



**Try to avoid**
Gridlines and an axis to read values from.

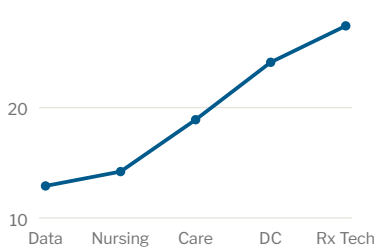



**Recommended**
Values on the bars; no axis, no grid.

## Zero baseline




**Try to avoid**
A truncated axis exaggerates differences.




**Recommended**
A line or dot plot can use a narrow range.

# **Bar chart** best practice, continued

Four more, from reviewing workforce decks.

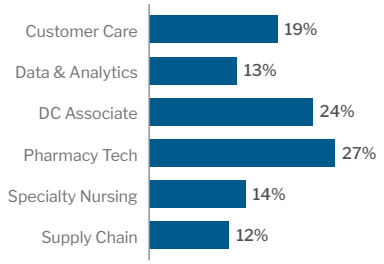
**Sort with intent.** Alphabetical order hides the ranking. Sort by value unless the categories have a natural order: tenure bands, job levels, survey scales.

**Horizontal for long names.** Rotated labels are slow to read. Job-family names need horizontal bars.

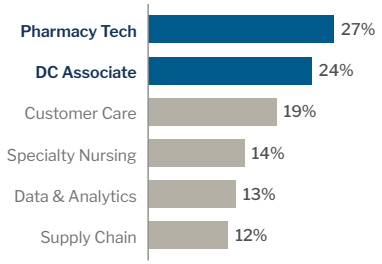
**Label the benchmark in place.** A reference line explained in a legend makes the reader search. Name it on the line.

**Two highlights at most.** Five colored bars in two colors is decoration, not emphasis.

## Sort with intent

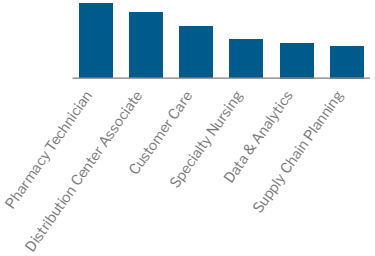



**Try to avoid**
Alphabetical: the ranking is hidden.

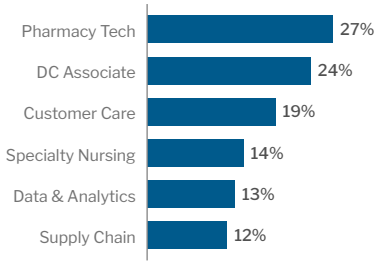



**Recommended**
By value, largest on top.

## Horizontal for long names

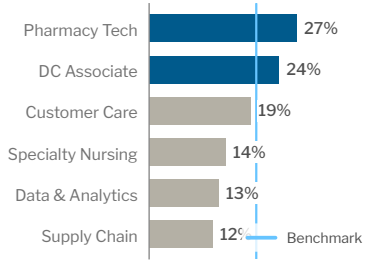



**Try to avoid**
Tilted names the reader has to decode.

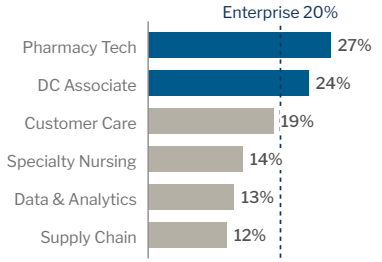



**Recommended**
Horizontal bars; names read naturally.

## Label the benchmark

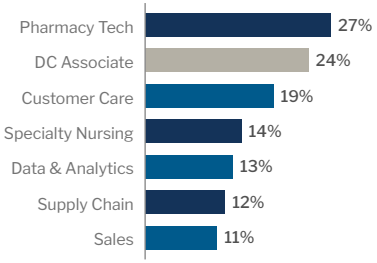



**Try to avoid**
A legend for one line, in a highlight color.

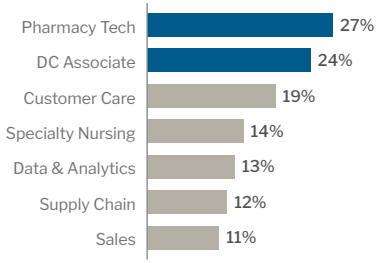



**Recommended**
Navy dashed line, named on the line.

## Two highlights at most




**Try to avoid**
Everything highlighted means nothing is.




**Recommended**
The two the headline names.

# **Build** a bar chart

The same chart in every tool we use. Each recipe starts from the starter kit, so the colors and fonts are already right.

Full, tested code for each platform is on the cookbook page: docs/cookbook/bar.md.

### Before you send it

- [ ] Starts at zero; bar about 2× the gap
- [ ] Sorted by value or a natural order
- [ ] Values labelled; no value axis or gridlines
- [ ] At most two highlights, matching the headline
- [ ] Benchmark labelled in place
- [ ] No group under 10 people

| | |
|---|---|
| **PowerPoint and Excel** | Start from the starter deck or chart workbook in `dist/office/`. Insert a clustered bar, sort the data ascending (so the largest is on top), set **Gap Width 50%**, fill all bars with Context grey and the focus bars with Focus blue. Add data labels at Outside End; delete the value axis, gridlines, legend and chart title. The slide title is the headline. |
| **Power BI** | Import `mckesson-dataviz-theme.json` and sort the bars by the measure. Drive the highlight from a measure, not by clicking bars: a `Bar color` measure returns the focus blue when attrition is above 20% and context grey otherwise, applied through **Colors › fx › Field value**. Turn on data labels; turn off the X axis and gridlines. See docs/powerbi-setup.md. |
| **Python** | matplotlib with `mckviz`; Plotly uses the same helpers (`mv.pl_highlight`). |

```
bars = ax.barh(df.job_family, df.rate, height=0.67,
               color=mv.highlight(df.job_family, focus=focus))
mv.label_bars(ax, bars, fmt="{:.1f}%")
mv.clean_axes(ax, keep=("left",), grid=None); ax.set_xticks([])
mv.titles(fig, headline, subtitle); mv.source(fig, source)
mv.lint(fig)   # checks colors, zero baseline, labels vs. axis
```

| | |
|---|---|
| **HTML** | Copy `templates/html/chart-card.html`: headline, subtitle and source are HTML, colors come from CSS variables such as `var(--mck-chart-focus)`, and the chart is drawn after the font loads. |
| **R** | `source("dist/r/theme_mck.R")`, then `geom_col(aes(fill = job_family), width = 0.67)` + `scale_fill_manual(values = mck_highlight(levels(df$job_family), focus = focus), guide = "none")` + `theme_mck()`. |
| **Coding agents** | "Make a McKesson bar chart of voluntary attrition by job family, FY26, highlighting the families above the enterprise rate." The `mck-chart` skill runs the workflow and returns the chart, a chart brief and alt text. |

PART-TO-WHOLE

# **Stacked** bar chart

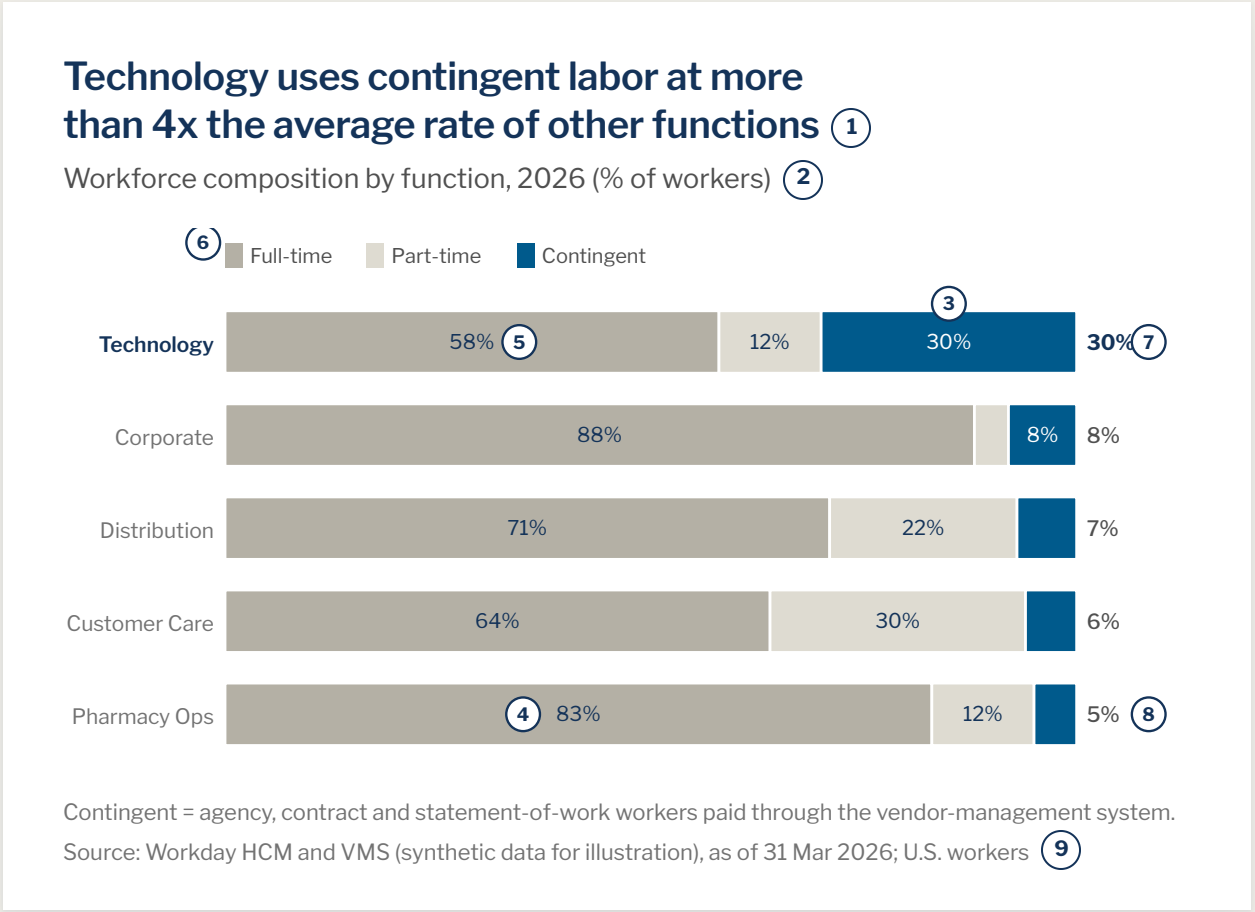
Part-to-whole charts show how a whole breaks into its parts, and how each part relates to the total.

A 100% stacked bar compares the mix of several wholes at once. Color only the segment the headline is about, and put it on the baseline or at the end of each bar.

### McKesson spec

- Five segments at most; group the rest as "Other"
- White separators between segments
- Labels inside segments that fit; key top-left in stack order
- Sort the bars by the segment that matters

1. **Insight headline**
   Names the part and how far it stands out.
2. **Subtitle**
   Metric, unit, scope, period.
3. **The part that matters**
   One segment in blue, at the end of every bar.
4. **Grey context**
   Two greys for the other parts.
5. **Labels in the segments**
   Where they fit; navy on grey, white on blue.
6. **Key in stack order**
   Top-left, left to right as the segments run.
7. **Echo the value**
   The story segment's value at the bar end, bold for the focus bar.
8. **Sorted by the story**
   Bars ordered by the blue segment, not alphabetically.
9. **Footnote and source**
   Define the parts.



PART-TO-WHOLE

# **Stacked bar** variants

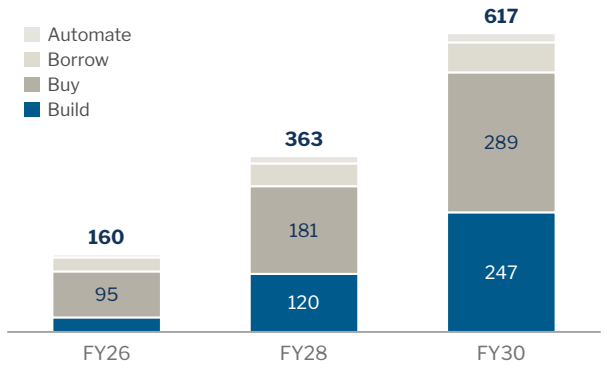
Same rules: one part in color, the rest grey, labels in place.

### When a stack is the wrong chart

- One whole with up to five parts: a **donut** or a single bar
- How a total changes step by step: a **waterfall**
- Many parts or a hierarchy: a **treemap**
- Comparing the parts themselves across groups: **grouped bars** or **small multiples**; only the baseline segment of a stack compares cleanly

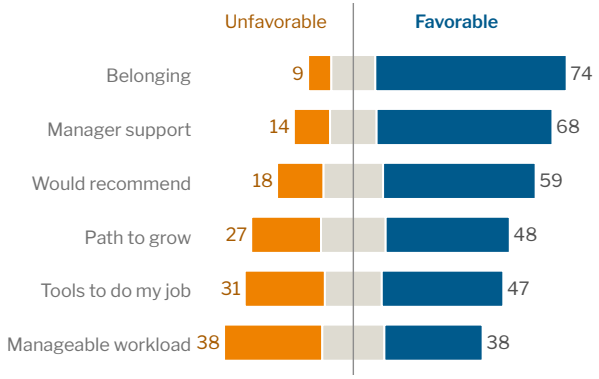
### Stacked columns with totals

When the total matters too. The FY30 Data & AI gap of 617 FTE: reskilling (on the baseline) closes 40%.



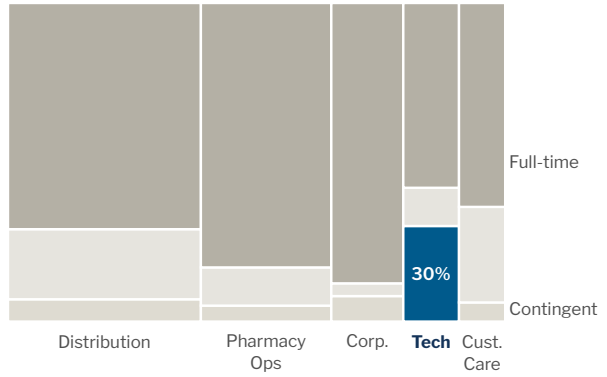

### Diverging stacked bars

Survey scales. Unfavorable in orange left of zero (it means worse), favorable in blue, neutral split across the middle.



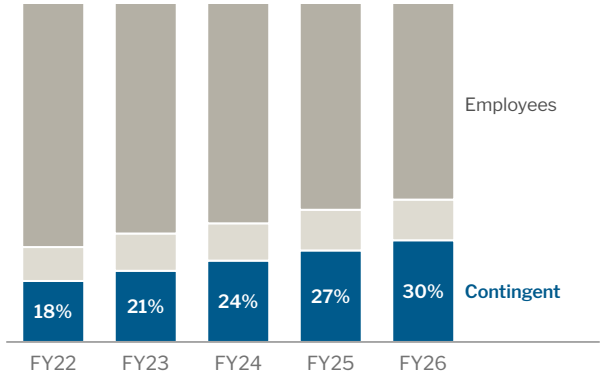

### Marimekko

Width and height both carry data: column width is each function's workforce, height its mix. Powerful, but it needs explaining.




### Mix over time

A few periods as 100% columns, the tracked segment on the baseline so its change reads directly.



PART-TO-WHOLE

# **Stacked bar** best practice

Stacks are easy to make and easy to misread. Four rules keep them honest.

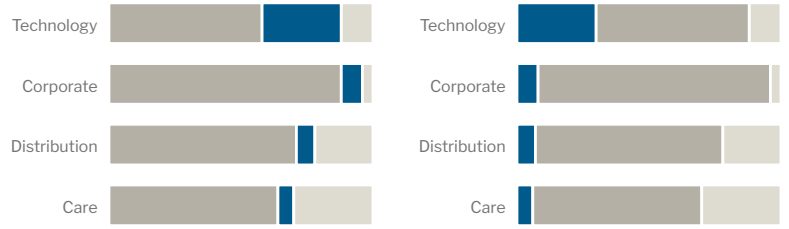
**The part that matters on the baseline.** Only segments that share a starting line can be compared by eye.

**Five segments at most.** More slices turn a bar into a barcode. Keep the parts the story needs and group the rest.

**Label in place; key in stack order.** If a key is needed, put it top-left, in the same order as the segments.

**One order in every bar.** Never re-sort segments within each bar.

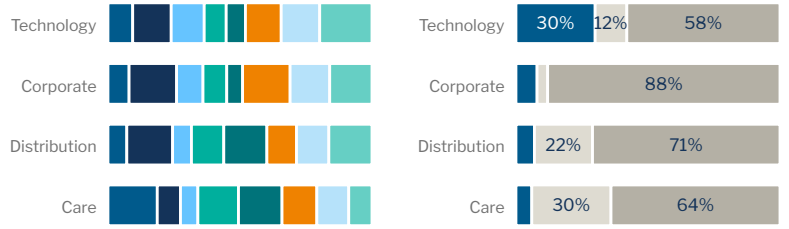
## The part that matters on the baseline




**Try to avoid**
A floating segment can't be compared across bars.

**Recommended**
On the baseline, every bar starts level.

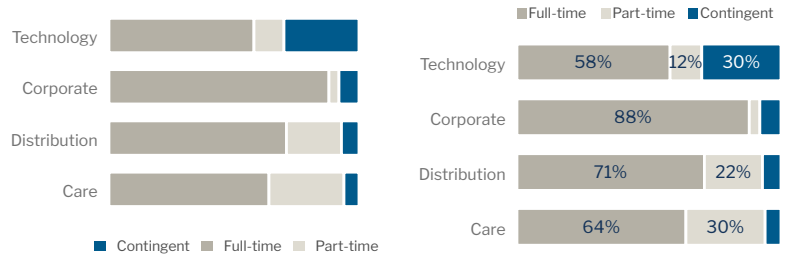
## Five segments at most




**Try to avoid**
Eight colored slices nobody can follow.

**Recommended**
The part in blue, the rest in grey.

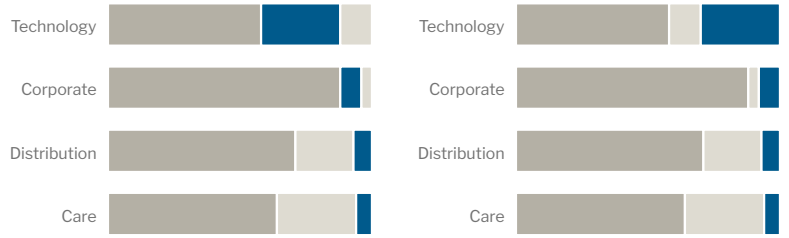
## Label in place, key in order




**Try to avoid**
A bottom legend in a different order.

**Recommended**
Values inside, key top-left in stack order.

## One order in every bar




**Try to avoid**
Segments re-sorted inside each bar.

**Recommended**
The same order everywhere.

PART-TO-WHOLE

# **Waterfall** chart

A waterfall shows how a starting total becomes an ending total, one driver at a time.

It is a stacked column chart with an invisible spacer series that floats each driver at the running total. Use it for labor cost bridges, headcount bridges and budget-to-actual walks.

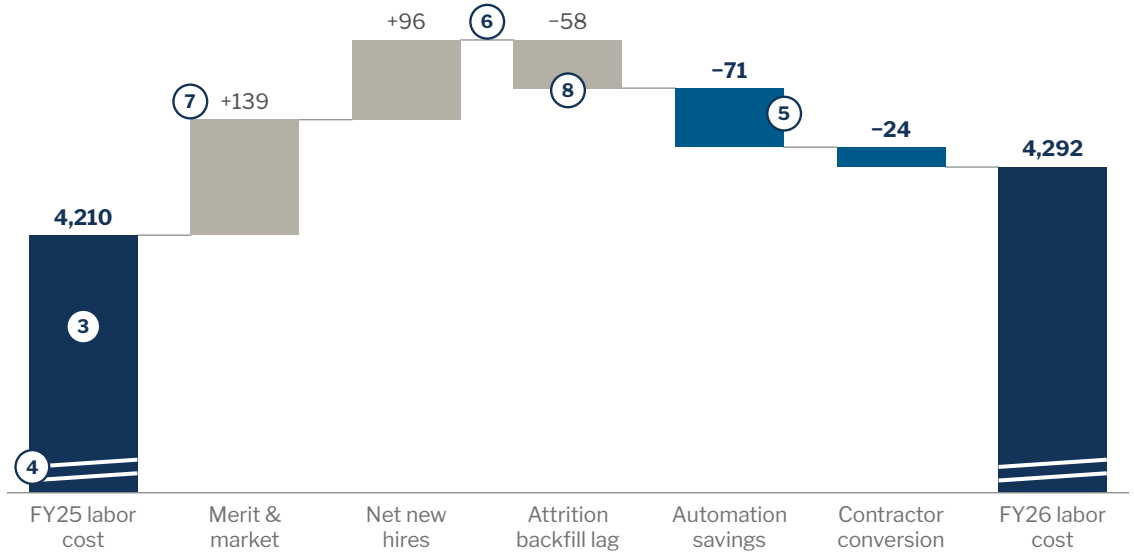
### McKesson spec

- Totals in navy; drivers grey; the drivers the headline names in blue
- Thin light connectors between bars
- Signed labels on drivers (true minus, explicit +); bold totals
- If the axis starts above zero, mark the break and say so

1. **Insight headline**
   Sizes what the named drivers did to the total.
2. **Subtitle**
   From and to, with the unit.
3. **Totals in navy**
   Start, end and any subtotal.
4. **Marked break**
   The axis starts at $3,900M; white slashes say so.
5. **The drivers that matter**
   In blue; label bold.
6. **Thin connectors**
   Light grey, so the bars stay dominant.
7. **Signed labels**
   +139, −58: direction without color.
8. **Grey context**
   Every other driver, in logical order.
9. **Footnote**
   States the axis break.

## Automation and contractor conversion offset ~40% of FY26 labor cost growth (1)

Labor cost bridge, FY25 to FY26 ($M) (2)




Value axis starts at $3,900M (break marks on the totals) so the drivers are readable.

Source: Finance labor cost model (synthetic data for illustration), FY26 (9)

PART-TO-WHOLE

# **Waterfall** variants

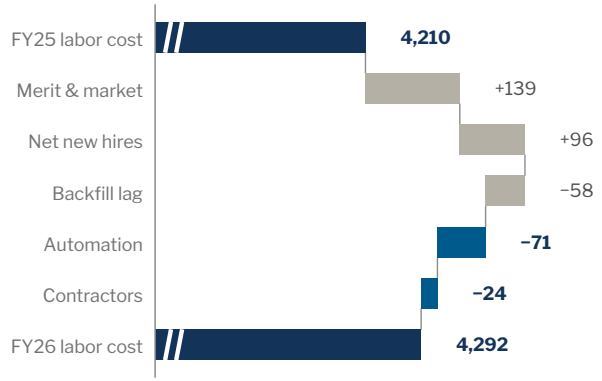
The same build, turned or recolored for the job.

**Color by direction only when direction is the point.** In a headcount bridge, additions in blue and losses in orange read instantly, because orange means "worse". In a cost bridge the default stays grey with a blue highlight.

Excel's built-in Waterfall chart is vertical only. For horizontal waterfalls or per-bar colors, build a stacked bar with a transparent spacer series (see the cookbook).

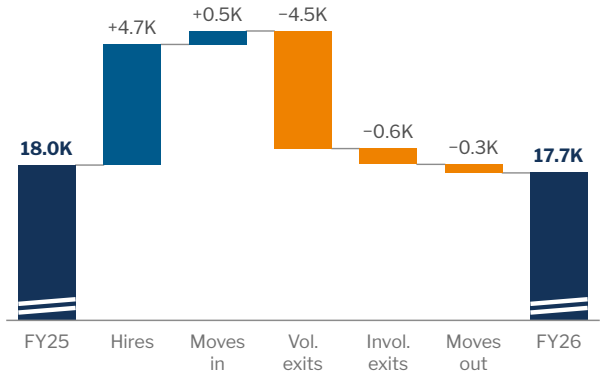
### Horizontal waterfall

Long driver names, narrow layouts and phones.



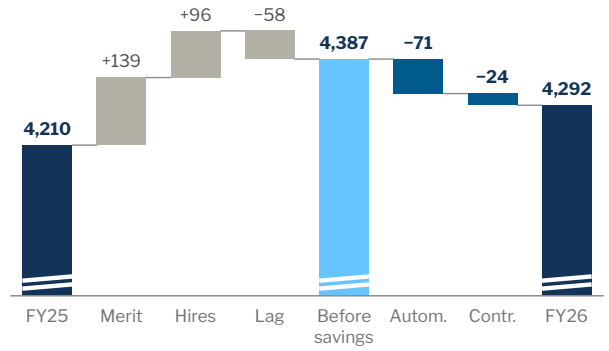

### Colored by direction

Frontline headcount bridge: voluntary exits equal 97% of external hires.



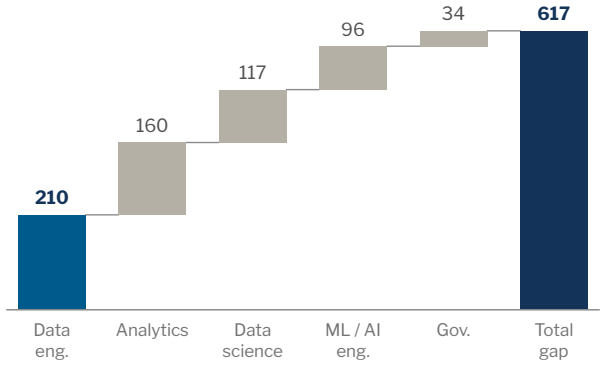

### With a subtotal

A light-blue subtotal separates cost growth from the savings that offset it.




### Distribution waterfall

Contributions to one total instead of steps in time: data engineering is 34% of the FY30 gap.



PART-TO-WHOLE

# **Waterfall** best practice

The bars carry the story; everything else stays quiet.

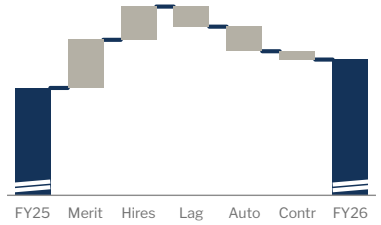
**Thin connectors.** Heavy dark connectors compete with the bars. Use hairlines in light grey.

**No red and green.** They fail for color-blind readers and aren't McKesson colors. Grey drivers with a blue highlight, or blue and orange when direction is the story.

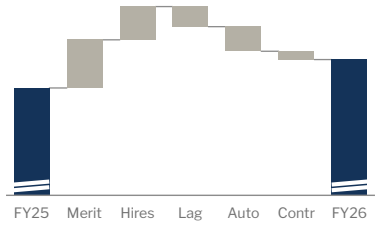
**Mark a broken axis.** Totals often dwarf the drivers. Starting the axis above zero is allowed here, and only here, if the break is drawn and footnoted.

**Group the small steps.** Keep the drivers the story needs; fold the rest together.

## Connector lines

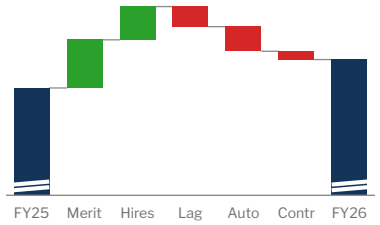



**Try to avoid**
Heavy, dark connectors.

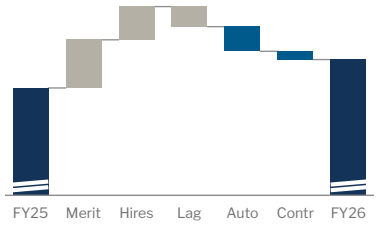



**Recommended**
Thin, light grey.

## Coloring the bars

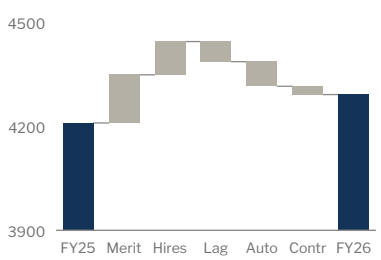



**Try to avoid**
Green up, red down.

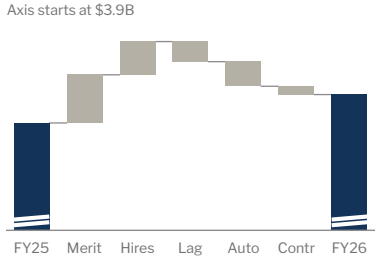



**Recommended**
Grey drivers; the named ones blue.

## A broken axis

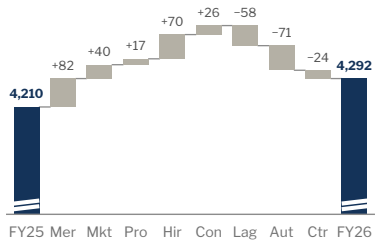



**Try to avoid**
Cut totals with no warning.

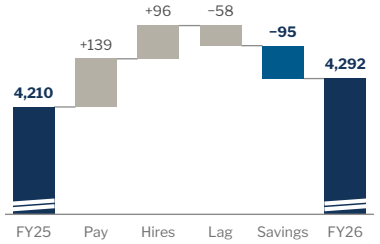



**Recommended**
Break marks and a note.

## Group the small steps




**Try to avoid**
Ten narrow drivers to decode.




**Recommended**
Five steps, savings combined.

PART-TO-WHOLE

# Treemap

Nested rectangles whose area is proportional to value. A treemap shows many parts, or a hierarchy, in one compact block.

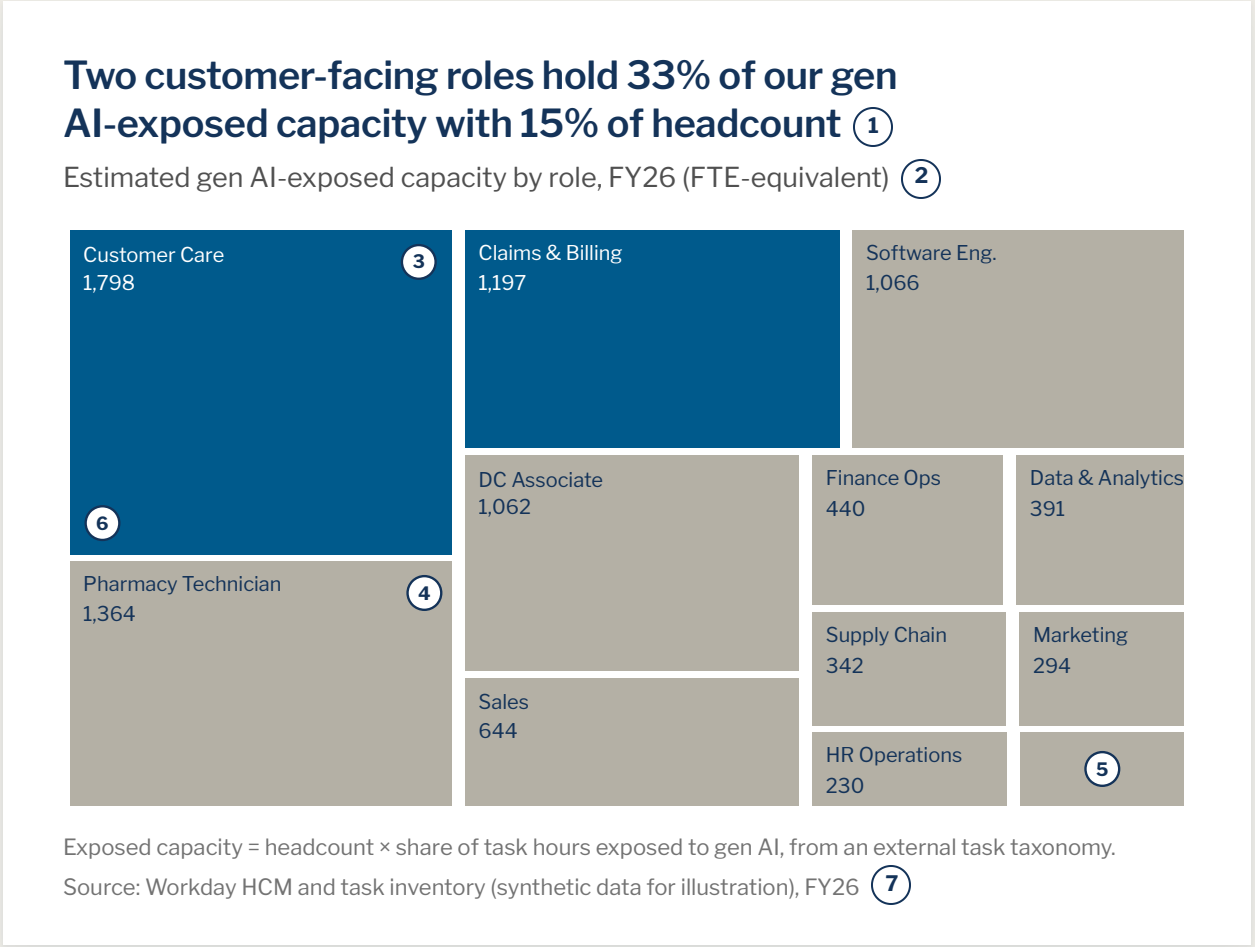
Use it when there are too many parts for a donut or a stack (ten to thirty roles, sites or skills) and the point is which few dominate.

Don't use it when readers need to compare similar sizes precisely, or there are only a handful of parts: a sorted bar is clearer.

### McKesson spec

- Squarified layout, largest top-left
- Thin, even white gutters
- Grey tiles; the ones the headline names in blue
- Name and value inside each tile that fits

1. **Insight headline**
   Share of the whole held by the named few.
2. **Subtitle**
   What area measures, with the unit.
3. **Focus in blue**
   The two roles the headline names.
4. **Grey context**
   Every other role, the same grey.
5. **Smallest last**
   Bottom-right. Tiles too small for a label get none; list them in a footnote if they matter.
6. **Largest first**
   Top-left, flowing to the smallest; labels top-left inside each tile.
7. **Footnote and source**
   Define the measure.



PART-TO-WHOLE

# **Treemap** best practice

A treemap works when it reads as one shape that flows from large to small.

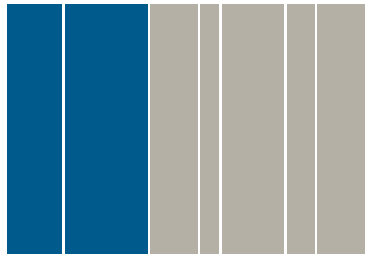
**Largest to smallest.** Don't arrange tiles alphabetically or by an org grouping that breaks the flow.

**Labels inside.** No label bands or boxes over the tiles; they distort the areas. If a tile can't hold its label, leave it unlabelled or label it outside.

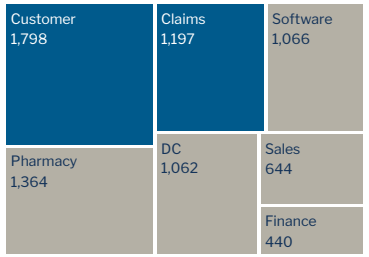
**Even padding.** Uneven gaps make similar tiles look different.

**Few parts? Use a bar.** Areas are harder to compare than lengths.

## Arrange by size

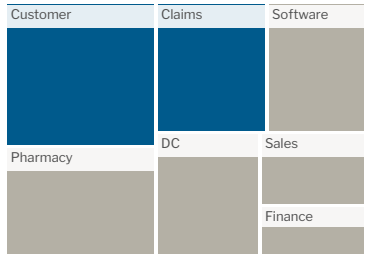


**Try to avoid**
Alphabetical strips.

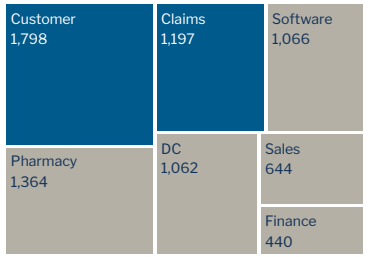



**Recommended**
Largest top-left.

## Labelling




**Try to avoid**
Label bands over the tiles.




**Recommended**
Inside, top-left.

## Padding

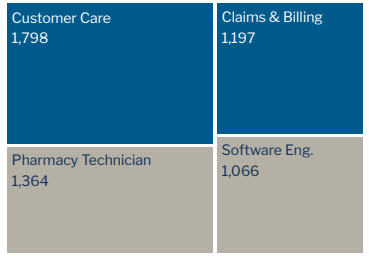


**Try to avoid**
Uneven gaps.

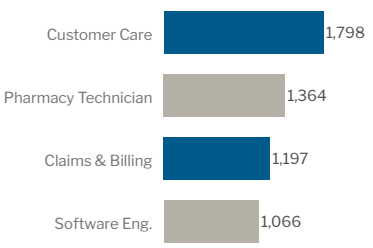


**Recommended**
Thin, even gutters.

## Few parts




**Try to avoid**
Four similar tiles.




**Recommended**
A sorted bar ranks them.

PART-TO-WHOLE

# **Donut** and pie

A donut shows one whole split into a few parts. Prefer it to a pie: readers follow the arc length around the ring, and the center holds the total.

Use it for one whole and at most five parts. To compare several wholes, use 100% stacked bars; for more parts, a bar or treemap.

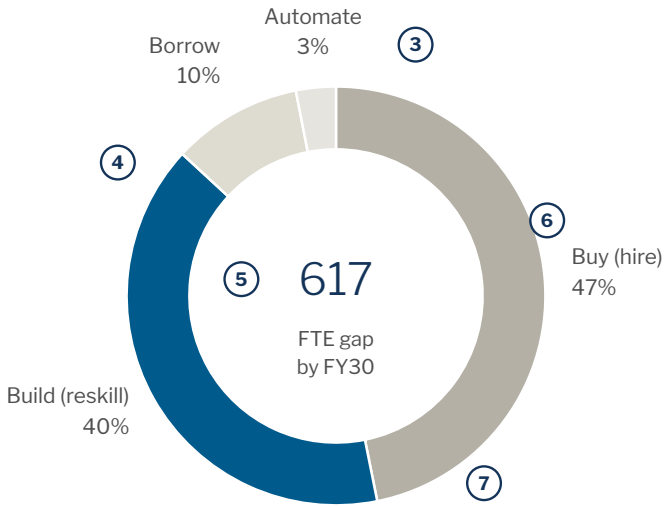
### McKesson spec

- Five segments at most; "Other" last
- Largest first, clockwise from 12 o'clock
- Hole about 60% of the diameter; white separators
- Labels and values outside; no legend

1. **Insight headline**
   The share, and what it means for the plan.
2. **Subtitle**
   The whole, the unit, the year.
3. **Start at 12**
   Largest first, then clockwise.
4. **The part that matters**
   In blue; the rest grey.
5. **Total in the center**
   Light weight, large.
6. **Labels outside**
   Name and share, beside each segment.
7. **White separators**
   Thin gaps keep segments distinct.
8. **Footnote and source**
   Define the parts.

## Reskilling can close 40% of the FY30 Data & AI gap; hiring still has to cover nearly half (1)

Planned sources to close the Data & AI capability gap, FY30 (% of 617 FTE) (2)




Base scenario. Build = reskilling current employees; buy = external hires; borrow = contractors and partners; automate = roles removed by automation.

Source: Strategic workforce planning model (synthetic data for illustration), Mar 2026 (8)

PART-TO-WHOLE

# **Donut** best practice

People judge length faster than angle or area, which is why the donut beats the pie.

**Donut over pie.** The ring turns angles into arc lengths, and the hole holds the total.

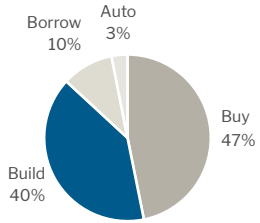
**Labels outside.** Squeezed labels on the ring are hard to read; put names and values outside.

**Few segments.** Group the smallest into "Other".

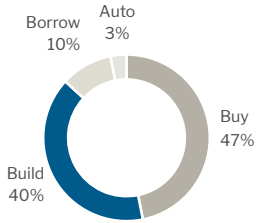
**Largest first, clockwise from 12.** Don't mix small and large at random.

Progress rings and dials for a single percentage are in Single number.

## Donut over pie

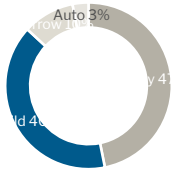



**Try to avoid**
Angles and areas.

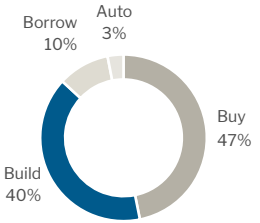



**Recommended**
Arc lengths on a ring.

## Labelling




**Try to avoid**
Labels squeezed onto the ring.

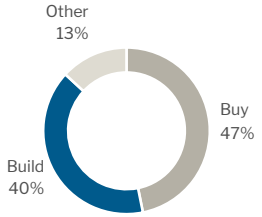



**Recommended**
Outside, with the share.

## Many small segments

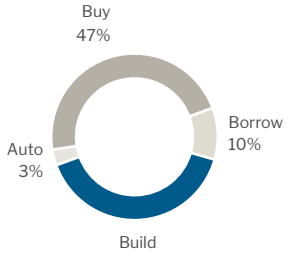


**Try to avoid**
Nine slices, nine colors.

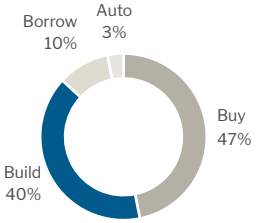



**Recommended**
The story, plus "Other".

## Order of segments




**Try to avoid**
No order, odd start.




**Recommended**
Largest first from 12.

PART-TO-WHOLE

# **Build** part-to-whole charts

Every tool can make these natively. The settings below are the ones that matter; tested code and click-by-click steps are on the cookbook pages.

stacked-bar.md · waterfall.md · treemap.md · donut-pie.md

### Before you send it

- [ ] Parts sum to the whole (100%, or the end total)
- [ ] Five parts at most; "Other" last
- [ ] One part in color, matching the headline
- [ ] Waterfall axis break marked and footnoted
- [ ] No group under 10 people, in any part

| | |
|---|---|
| **PowerPoint and Excel** | **Stacked:** Stacked or 100% Stacked Bar, Gap Width 50%, Overlap 100%, white 1.5 pt borders; move the story series first or last in Select Data. **Waterfall:** Insert › Waterfall, then right-click each total › *Set as Total*; drivers grey, totals navy. **Treemap:** Insert › Hierarchy › Treemap; labels *Overlapping*, not Banner. **Donut:** Doughnut, angle of first slice 0°, hole 62%, data sorted descending. |
| **Power BI** | Native 100% stacked bar, Waterfall, Treemap and Donut visuals all take the theme. Set the story color per series or with a field-value color measure; legend top-left or off; data labels on. The native waterfall can't color single bars or break the axis: use a stacked column with a transparent spacer measure when you need that. |
| **Python** | `ax.barh(..., left=running_total, edgecolor="white")` for stacks; the waterfall, squarified treemap and donut recipes in the cookbook use only matplotlib and `mckviz`. Plotly has native `go.Waterfall`, `go.Treemap` and `go.Pie(hole=0.6)`. |
| **HTML** | Plotly with `MCK_PLOTLY_TEMPLATE`: `barmode: "stack"`, `waterfall` and `treemap` traces. Read colors from `var(--mck-chart-focus)` and `var(--mck-chart-context)`. |
| **R** | `geom_col(position = "fill")` for 100% stacks; `treemapify::geom_treemap()` for treemaps; waterfalls with `geom_rect()`. Colors from `mck_highlight()`. |

```
colors = mv.highlight(parts, focus="Contingent")          # one part blue, rest grey
mv.lint(fig)    # flags > 6 colors, legends, and bars that don't start at zero
```

CHANGE OVER TIME

# **Line** chart

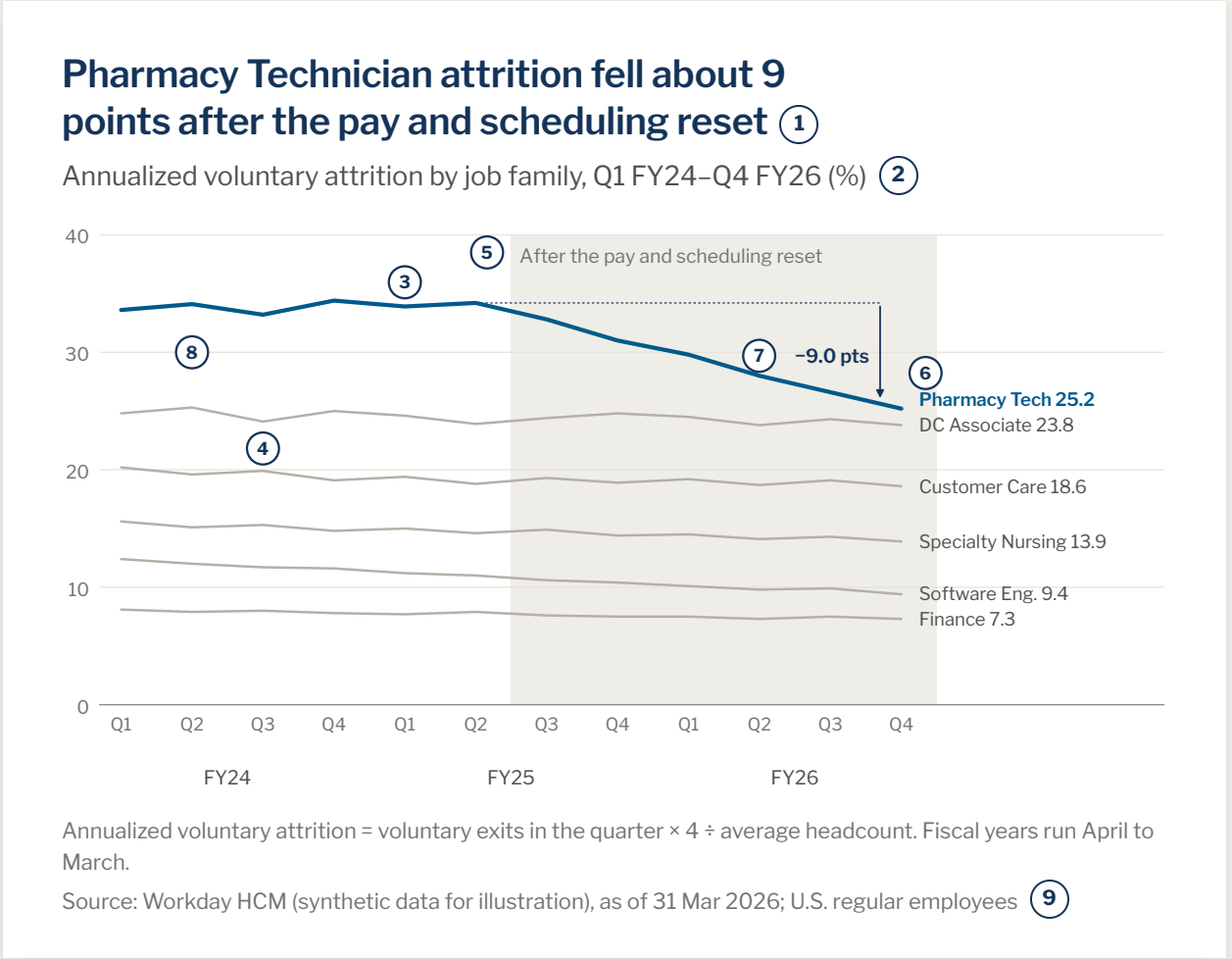
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

### McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHANGE OVER TIME

# Line chart variants

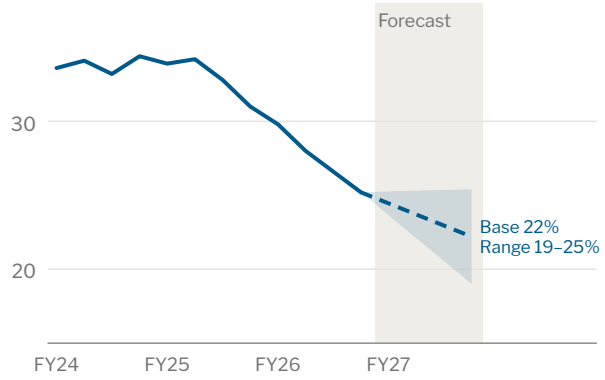
The same rules hold: one line in blue, names at the ends, light scaffolding.

### When a line is the wrong chart

- Only two to five periods, and the amounts matter: **columns**
- How a total splits over time: **stacked area**
- Discrete events of different sizes: a **bubble timeline**
- Many series that tangle: **small multiples**
- Two measures in different units: two aligned charts, never a **dual axis**

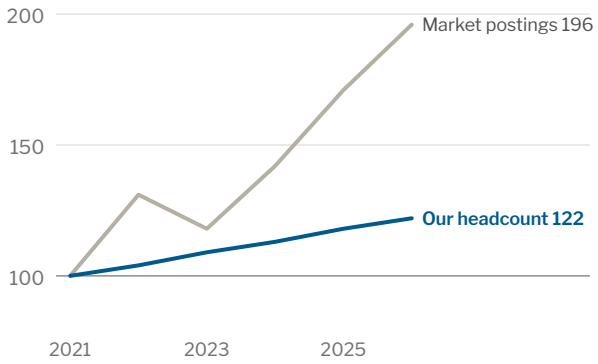
### Forecast with a range

Dashed beyond the last actual, inside a band that widens with time. Label it as an estimate.



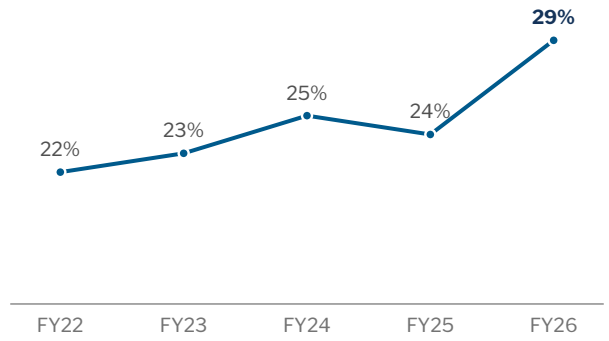

### Indexed to 100

Compares growth when the units or sizes differ: market Data & AI postings reached 196, our headcount 122.



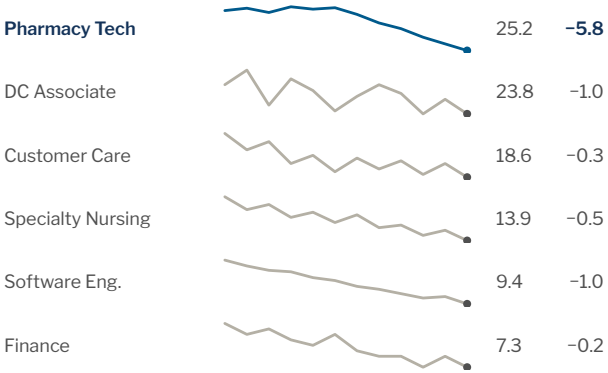

### Few points, every value labelled

With five annual values, label each one and drop the axis.




### Sparklines

One small line per row in a table or dashboard: shape, latest value, change. Worse in orange-brown.



CHANGE OVER TIME

# **Line chart** best practice

Four rules that make trends readable.

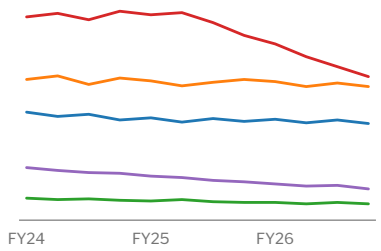
**Highlight one line.** A color per series doesn't show the trend; it makes the reader decode. Grey the background, color the message.

**Lines for trends.** Stacked columns hide how each part changes. When the trend is the point, draw lines.

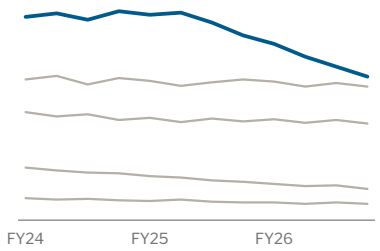
**No fill without a zero.** Shading under a line turns it into an area chart, which implies a zero baseline.

**Names at the line ends.** A legend above the chart breaks the reading order.

## Highlights

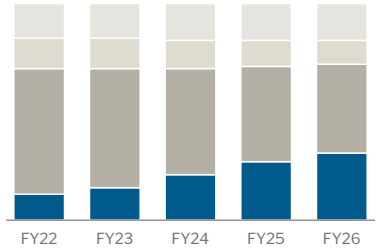



**Try to avoid**
Every line its own color.

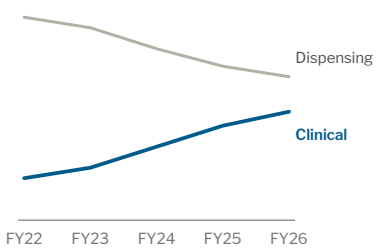



**Recommended**
Grey context, one blue line.

## Showing trends

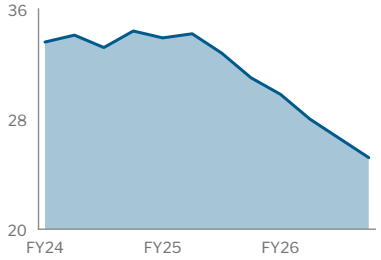



**Try to avoid**
Stacks hide each part's change.

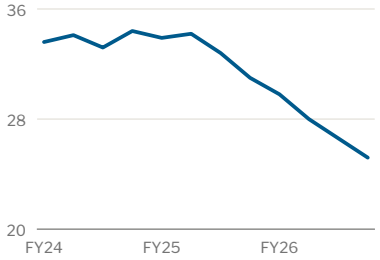



**Recommended**
Lines show the crossover.

## Shading lines

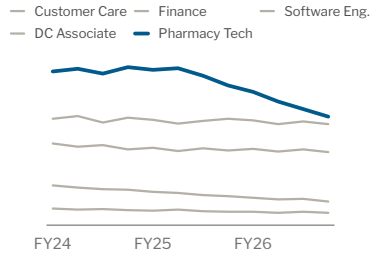



**Try to avoid**
A filled area on a 20–36 axis.

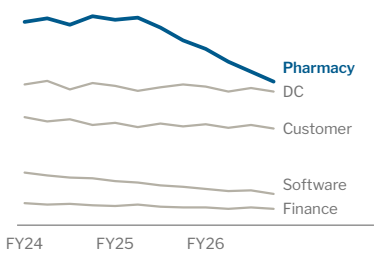



**Recommended**
A plain line.

## Labelling lines




**Try to avoid**
A legend to decode.




**Recommended**
Names where the lines end.

CHANGE OVER TIME

# **Line chart** best practice, continued

Four more, from HR and workforce data in particular.

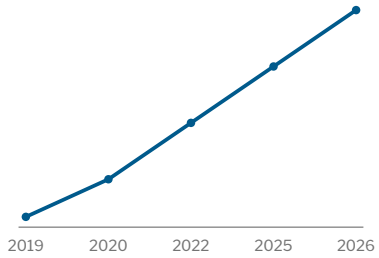
**Honest time spacing.** Missing years still take up space. Use a real date axis, not evenly spaced labels.

**Show gaps as gaps.** A system change or survey break is not a straight line. Leave the gap and say why.

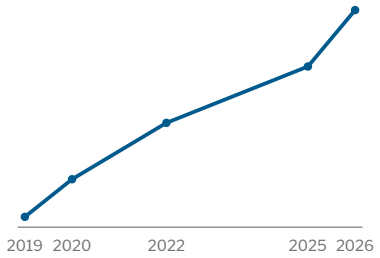
**Weight with intent.** Data lines about twice the gridlines; gridlines light, or gone.

**Too many lines? Split them.** More than four or five crossing lines become spaghetti. Use small multiples.

## Even time intervals

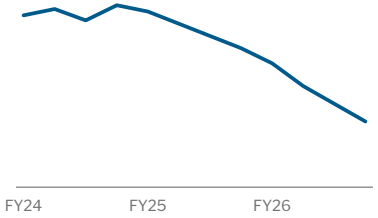



**Try to avoid**
Missing years squeezed out.

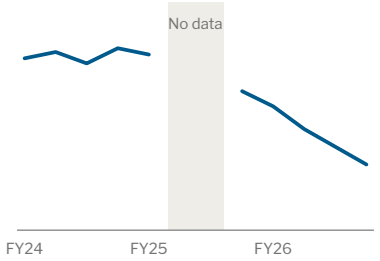



**Recommended**
True spacing; the slope is honest.

## Missing data

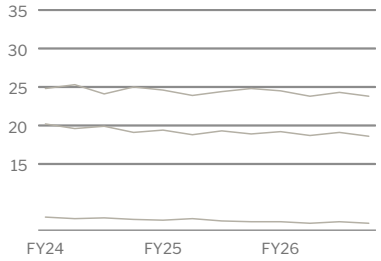



**Try to avoid**
Drawing across the gap.

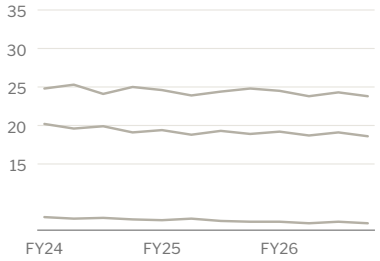



**Recommended**
A visible, labelled gap.

## Line weight

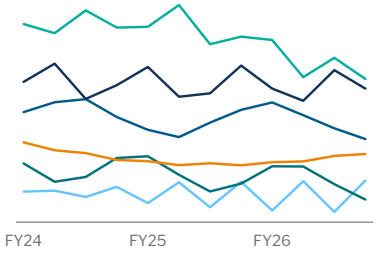



**Try to avoid**
Gridlines outweigh the data.

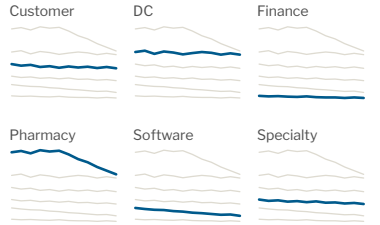



**Recommended**
Data lines twice the grid.

## Too many lines




**Try to avoid**
Six crossing colors.




**Recommended**
Small multiples.

CHANGE OVER TIME

# **Stacked** area chart

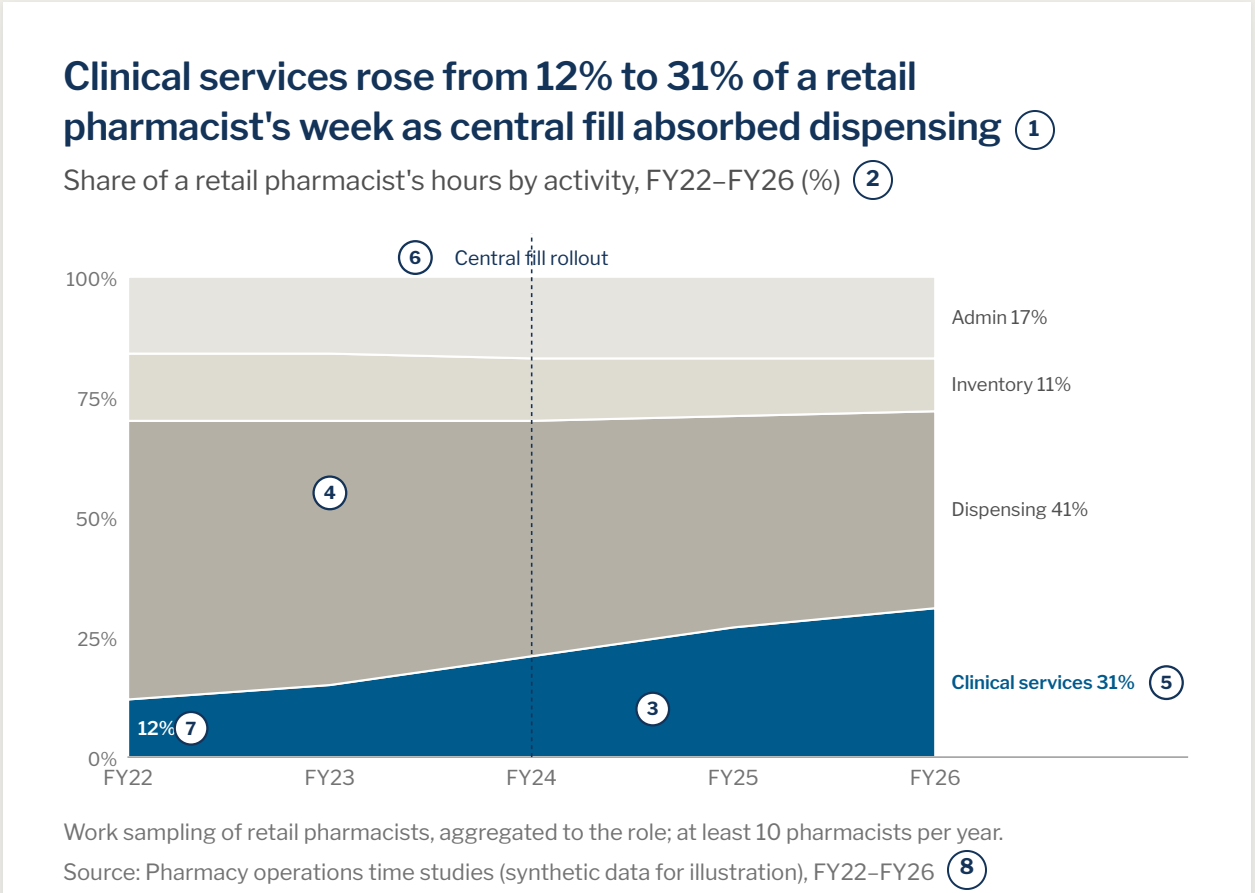
Layers stacked on each other show how a total splits over time, and how each part's share shifts.

Use it for a few parts that change gradually: how a role's hours divide between activities, or how the workforce mix moves. The layer on the baseline is the only one that reads precisely, so put the story there.

### McKesson spec

- Four or five layers at most
- The story layer on the baseline, in blue; the rest ordered greys
- Labels inside or at the right edge, not in a legend
- Mark the event that explains the shift

1. **Insight headline**
   Start, end and the reason.
2. **Subtitle**
   Whose time, what unit, which years.
3. **Story on the baseline**
   Blue, so its growth reads directly.
4. **Grey context**
   Ordered from dark to light.
5. **Labels at the end**
   Name and latest share beside each layer.
6. **The event, marked**
   A thin dashed line and a short name.
7. **Where it started**
   The first value, so the change is explicit.
8. **Footnote and source**
   How time was measured.



CHANGE OVER TIME

# **Stacked area** best practice

Areas are heavy marks. Keep them few, ordered and calm.

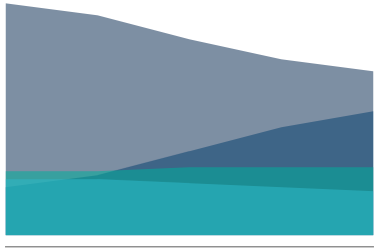
**Stack, don't overlap.** Areas drawn over each other hide what's behind. Stack them, or use lines.

**Order the colors.** Random colors make neighbors clash. Story layer blue on the baseline; the rest in ordered greys.

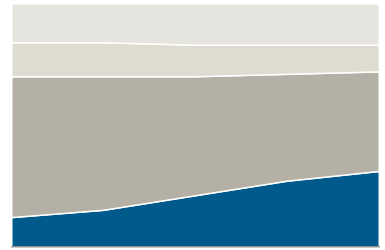
**Volatile on top.** A jumpy layer at the bottom makes every layer above it jump. Keep steady layers low.

**Few layers.** Four or five at most; fold the rest into "Other".

### Overlapping areas

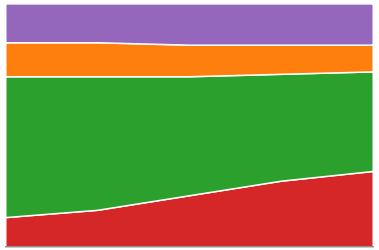


**Try to avoid**
Translucent layers over each other.

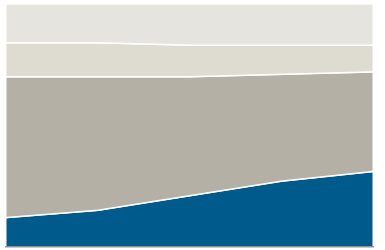


**Recommended**
Stacked layers.

### Color order



**Try to avoid**
Unrelated bright colors.



**Recommended**
Blue story, ordered greys.

### High variability

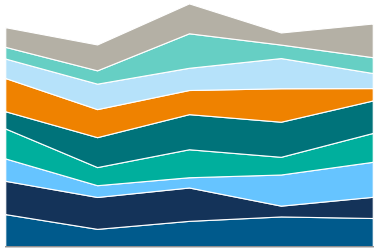


**Try to avoid**
The volatile layer at the bottom.



**Recommended**
Steady layers low, volatile on top.

### Number of layers



**Try to avoid**
Nine thin layers.



**Recommended**
Three: story, comparison, rest.

CHANGE OVER TIME

# **Bubble** timeline

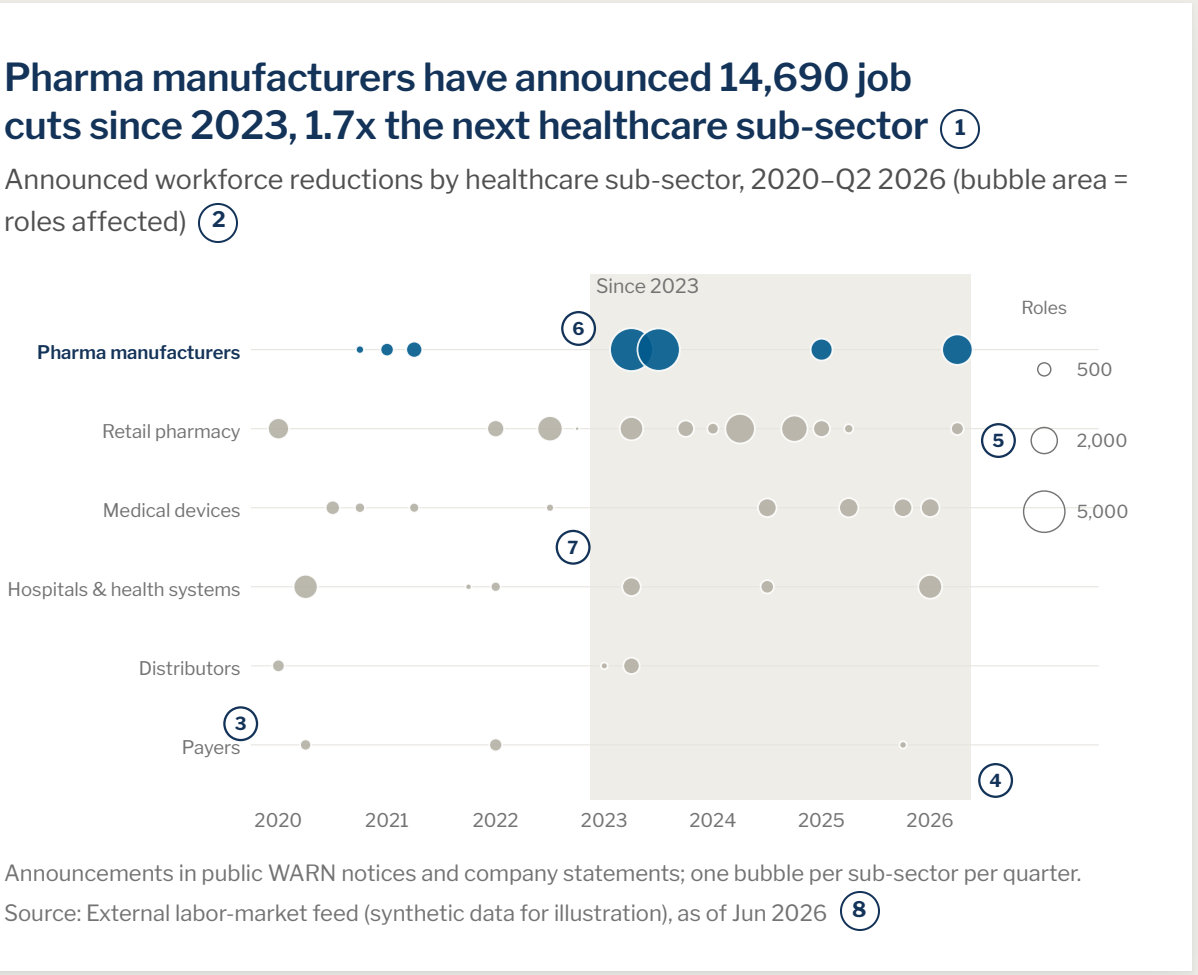
Events placed on a time axis, with bubble area showing size. It combines a timeline with a proportional-area chart.

Use it for discrete events of different sizes: announced layoffs in the talent market, reorganizations, hiring waves. Each category gets its own row.

### McKesson spec

- Bubble area, not diameter, proportional to value
- One grey for all rows; the headline's row in blue
- Largest bubbles drawn first, with white outlines
- A size key with two or three reference circles

1. **Insight headline**
   Who stands out, by how much.
2. **Subtitle**
   Says what bubble area means.
3. **One row per category**
   Sorted by total, largest on top.
4. **Time along the bottom**
   Even intervals, years labelled.
5. **Size key**
   Reference circles; area ∝ value.
6. **Focus row**
   Blue; white outlines keep overlaps readable.
7. **The period that matters**
   Shaded and named.
8. **Footnote and source**
   What counts as an event.



CHANGE OVER TIME

# **Bubble timeline** best practice

The rows already separate the categories; let size do the rest.

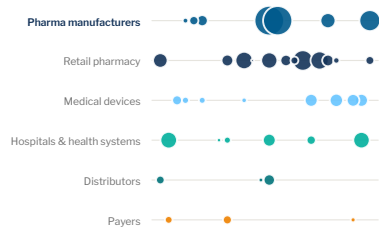
**One color.** Coloring each row repeats what the rows already say. Use grey, and blue for the headline's row.

**Don't hide small bubbles.** Draw the largest first, with white outlines.

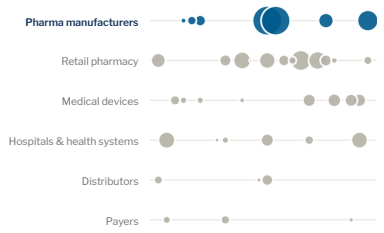
**Sort the rows.** By total, or by a meaningful order, not alphabetically.

**Show the scale.** Without a size key, readers can only say "bigger".

## Coloring bubbles

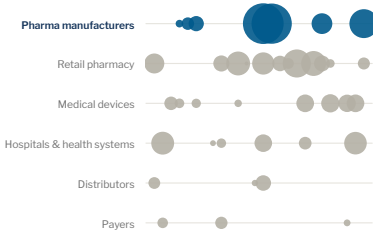



**Try to avoid**
A color per row.

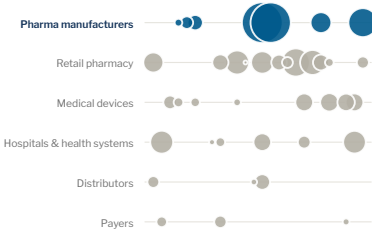



**Recommended**
Grey, one row blue.

## Overlapping bubbles

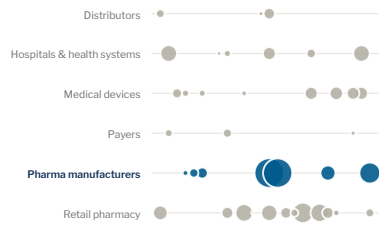



**Try to avoid**
Big bubbles on top.

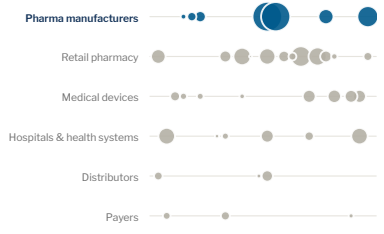



**Recommended**
Big behind, white outlines.

## Row order

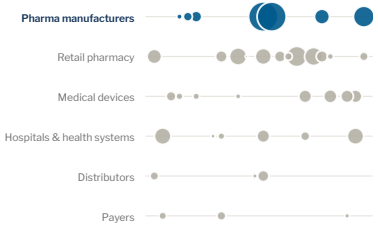



**Try to avoid**
Alphabetical rows.

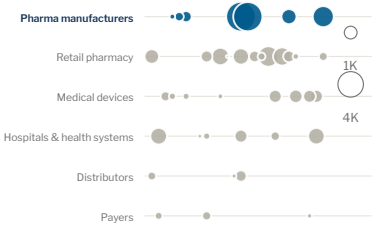



**Recommended**
Sorted by total.

## Size key




**Try to avoid**
No way to read size.




**Recommended**
Two reference circles.

CHANGE OVER TIME

# **Small** multiples

The same small chart repeated for each group, on one shared scale. The eye compares shapes across panels instead of untangling crossing lines.

Use it when six or more groups each need their own trend: business units, regions, job families. In each panel, the panel's own line is blue and the others sit faintly behind it.

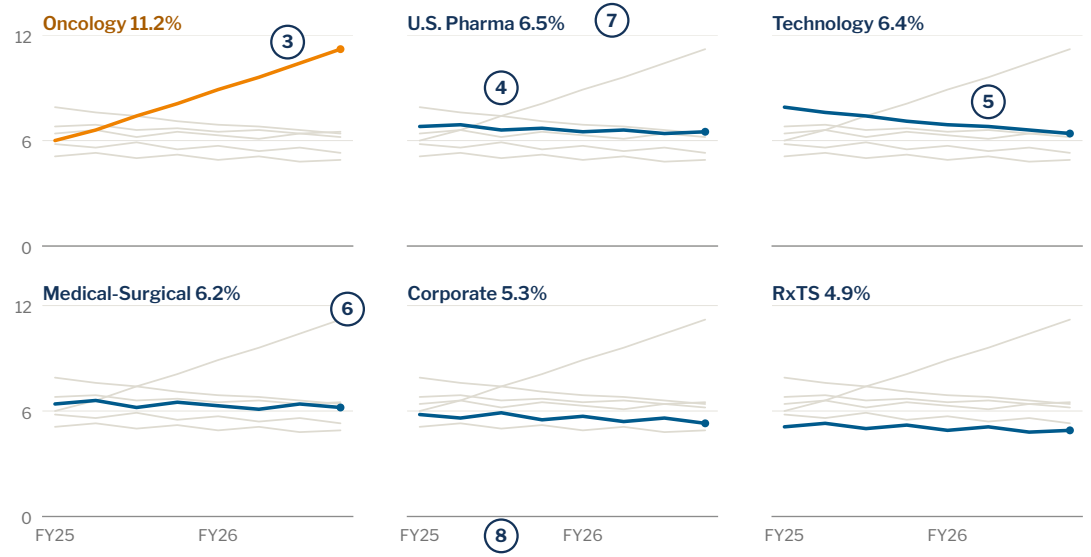
### McKesson spec

- One shared scale, stated once
- Panels ordered by value or a natural order
- Panel title = name + latest value
- Axis labels only on the outer edge

1. **Insight headline**
   The one panel that differs.
2. **Subtitle**
   Metric, unit, period, once for all panels.
3. **The exception**
   In orange, because it is getting worse.
4. **Its own line in blue**
   The other units faint behind it, for context.
5. **Ordered by value**
   Highest latest value first.
6. **Shared scale**
   0–12% in every panel.
7. **Title carries the number**
   Name and latest value.
8. **Outer axes only**
   Labels on the left column and bottom row.
9. **Footnote and source**
   Define the metric.

## Oncology regretted attrition nearly doubled in two years while every other business unit stayed within 2 points (1)

Regretted attrition by business unit, Q1 FY25–Q4 FY26 (% of high performers, annualized) (2)




Regretted attrition = voluntary exits of employees rated "exceeds" in the last review, annualized.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026 (9)

CHANGE OVER TIME

# **Small multiples** best practice

Small multiples only work if every panel is read the same way.

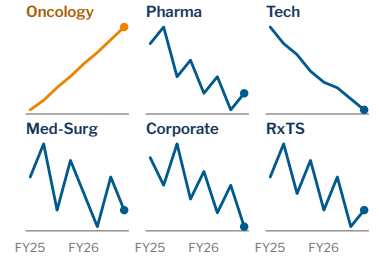
**One scale.** Independent scales make a 0.5-point wobble look like the 5-point rise.

**Context in every panel.** The other units, faint behind each line, show whether a panel is unusual.

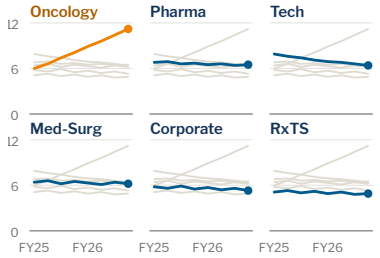
**Order with meaning.** By value, size or geography, not the alphabet.

**Label the edges only.** Repeated axes are clutter the eye reads six times.

## Shared scale

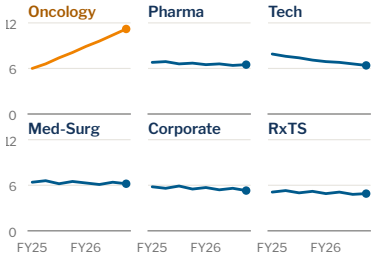



**Try to avoid**
Every panel zooms itself.

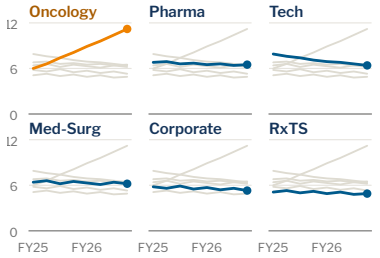



**Recommended**
One scale; the rise stands out.

## Context

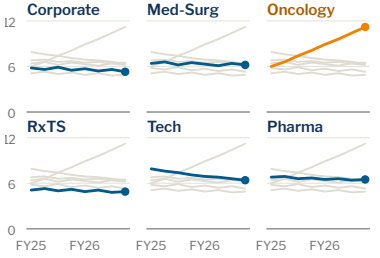



**Try to avoid**
Lines alone.

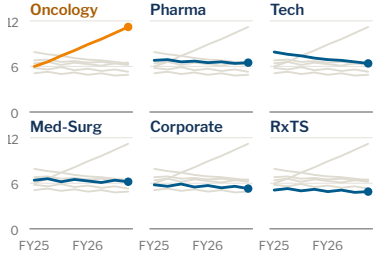



**Recommended**
The others faint behind.

## Panel order

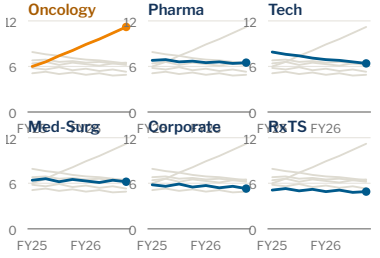



**Try to avoid**
Alphabetical.

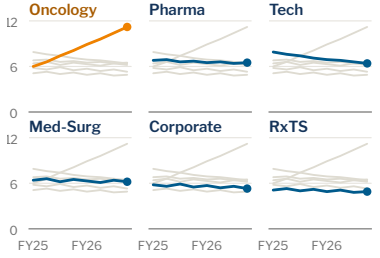



**Recommended**
By latest value.

## Axes




**Try to avoid**
Axes on every panel.




**Recommended**
Outer edge only.

CHANGE OVER TIME

# **Build** change-over-time charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

line.md · stacked-area.md · bubble-timeline.md · small-multiples.md

### Before you send it

- [ ] Time left to right, at true intervals
- [ ] One line or layer in blue, matching the headline
- [ ] Names at line ends; no legend
- [ ] Forecasts dashed, labelled and with a range
- [ ] Small multiples share one scale

| | |
|---|---|
| **PowerPoint and Excel** | **Line:** Insert › Line; format the focus series 2.25 pt Focus blue, the rest 1.25 pt grey; delete the legend and add data labels to the last point only (Series Name + Value). Use a *date* axis so missing periods keep their space. **Stacked area:** Insert › 100% Stacked Area; story series first in Select Data; white 1 pt borders. **Small multiples:** build one chart, fix the axis bounds (Minimum/Maximum, not Auto), then copy it per group. |
| **Power BI** | Line chart with the focus series colored by measure; turn off the legend and use *Series labels*. Forecast bands as low/high measures shown with *Error bars* › *Shaded area*. Small multiples are built in: add the group to *Small multiples* and keep *Y-axis* › *Shared*. |
| **Python** | matplotlib with `mckviz`; `mv.label_line_ends(ax)` places and nudges end labels; `mv.shade_period(ax, start, end, label)` marks the period. |

```
for fam, s in df.items():                # one column per job family, one row per quarter
    ax.plot(x, s, label=fam, color=C.focus if fam == focus else C.context,
            lw=2.4 if fam == focus else 1.4)
mv.shade_period(ax, 6, 11, "After the reset")
mv.label_line_ends(ax, fmt="{:.1f}")      # names at the ends, nudged apart
```

| | |
|---|---|
| **HTML** | Plotly `scatter` traces with `mode: "lines"`, an annotation per line end, and a `shape` for the shaded period; `facet_col` (Plotly Express) for small multiples with shared axes. |
| **R** | `geom_line()` with `mck_highlight()` colors and `geom_text()` end labels; `geom_area()` for stacks; `facet_wrap(~ unit)` keeps a shared scale by default. |

CORRELATION

# Bubble scatterplot

Correlation charts show how two or more measures relate to each other: does X go with Y, and who sits where it matters?

Each entity is one mark. Position carries two measures; bubble area can carry a third, such as headcount. Use it for exposure vs. pay by role, engagement vs. span of control by team, or time to fill vs. market tightness by metro.

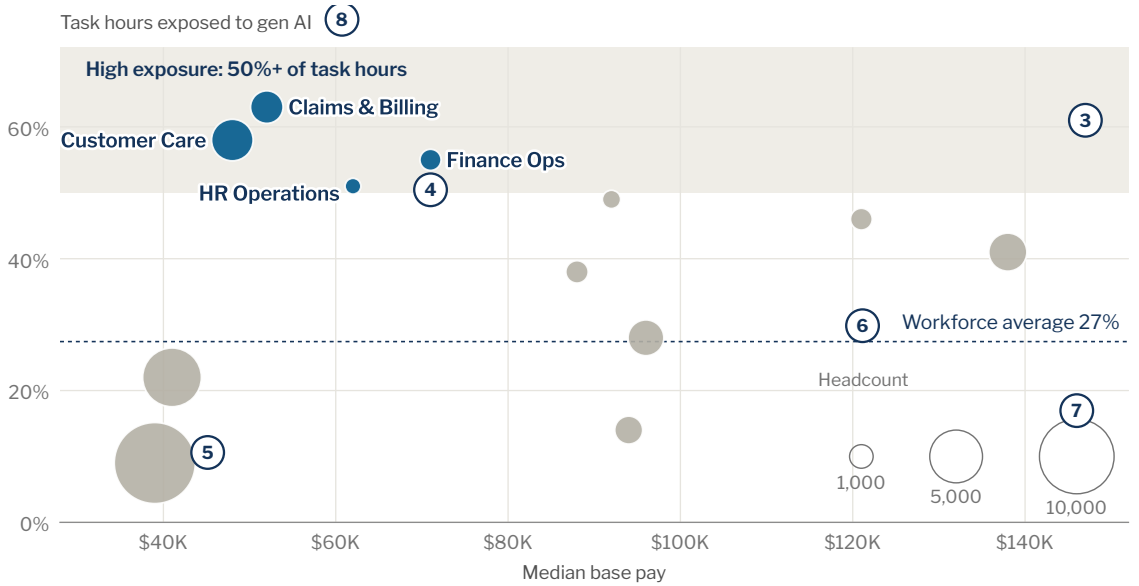
### McKesson spec

- Grey bubbles; the ones the headline names in blue
- Bubble area, not diameter, ∝ value; a size key
- Label five points at most; shade the zone that matters
- Say "is associated with", never "causes"

1. **Insight headline**
   Who is in the zone, and by how much.
2. **Subtitle**
   Both measures, the period, and what bubble area means.
3. **The zone that matters**
   Shaded behind the marks and named in place.
4. **Focus in blue, labelled**
   Only the bubbles the headline names.
5. **Grey context**
   Unlabelled; big bubbles drawn first, white outlines.
6. **Reference line**
   Navy, dashed, its value computed and labelled.
7. **Size key**
   Three round values on the same scale.
8. **Horizontal axis titles**
   Units on the axes; the Y title top-left.
9. **Footnote and source**
   Define the measure and the average.

## Four support roles have half or more of their task hours exposed to gen AI, 2x the 27% workforce average (1)

Share of task hours exposed to gen AI vs. median base pay, by role, FY26; bubble area = headcount (2)




Exposure = share of task hours matched to gen AI-capable activities in an external task taxonomy. Average weighted by headcount.

Source: Workday HCM and task inventory (synthetic data for illustration), FY26 (9)

CORRELATION

# Scatterplot variants

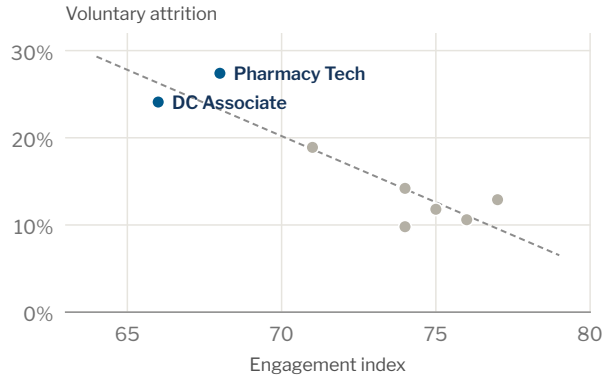
The same rules hold: grey context, blue for the story, labels only where the headline points.

### When a scatter is the wrong chart

- One measure, ranked: a **bar**
- One measure over time: a **line**
- The X measure is a category: a **dot plot**
- Thousands of overlapping people: a **binned heatmap**
- Four or more measures per entity: **parallel coordinates**

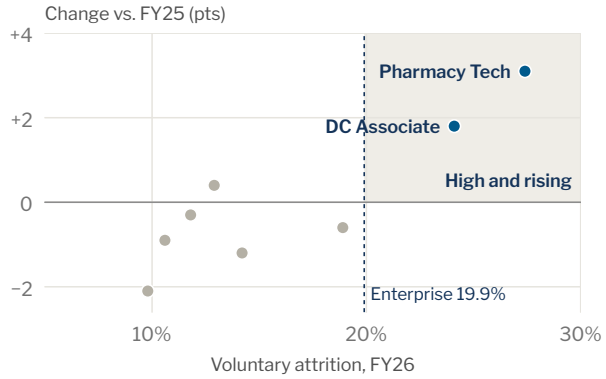
## Plain scatter

No size measure, equal dots. Lower engagement goes with higher attrition across job families (r = −0.91).



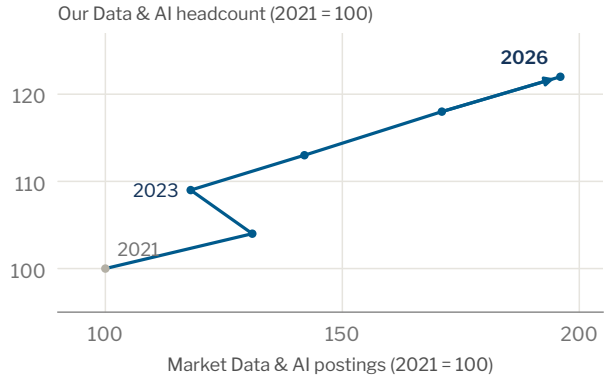

## Quadrant

Reference lines at a meaningful value; shade only the quadrant the headline names. Two families are high and rising.



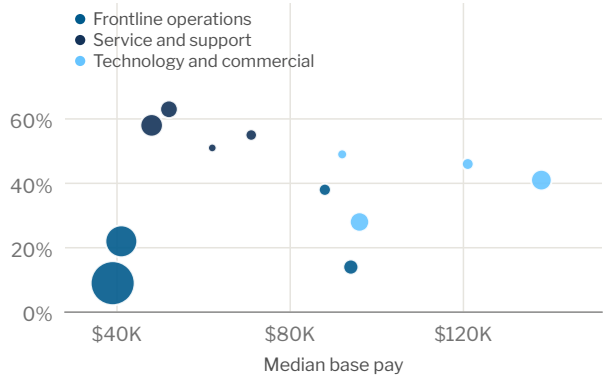

## Connected scatter

One entity over time, points joined in order. Market postings reached 196 while our headcount reached 122.




## Color for a real group

Color only when it encodes a variable, with a key: four groups at most, in palette order.



# **Scatterplot** best practice

A scatter invites the eye to roam. These four rules give it somewhere to land.

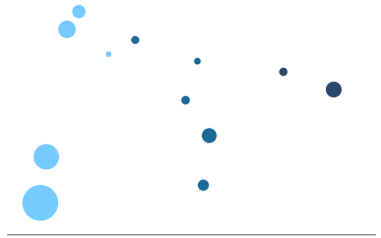
**Color carries a variable, or the story.** Coloring by x-range repeats the position and encodes nothing.

**Two trend lines at most.** One line per highlighted group; the rest stay grey, with no line.

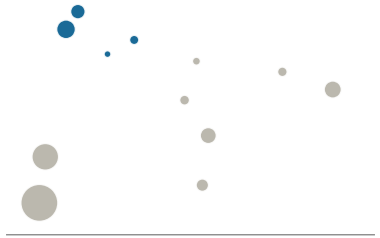
**Label what the headline names.** Twelve labels bury the point. Put the rest in tooltips.

**Big bubbles behind.** Draw the largest first, with white outlines, so small ones stay visible.

## Using color

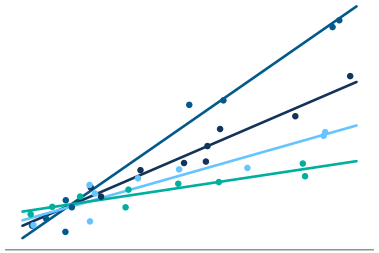


**Try to avoid**
Colors by pay range.

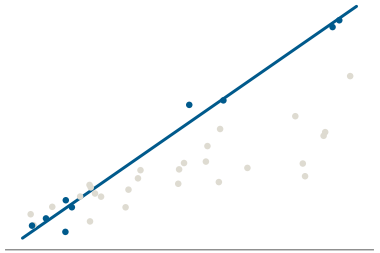


**Recommended**
Grey, the story in blue.

## Trend lines



**Try to avoid**
Four crossing lines.



**Recommended**
One group, one line.

## Labelling points




**Try to avoid**
Every bubble named.




**Recommended**
Only the focus.

## Overlapping bubbles



**Try to avoid**
The focus hidden underneath.



**Recommended**
Big first, white outlines.

CORRELATION

# Dual axis charts

Two measures in different units, one axis each. Both ranges are arbitrary, so where the lines cross or track is the designer's choice, not the data's.

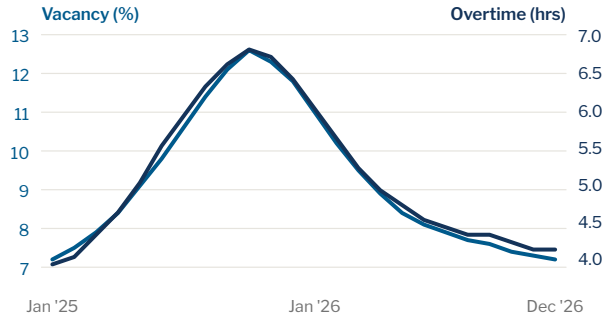
### Pick the alternative from the question

- Did B move when A moved? **Two aligned panels** on one time axis
- Which grew faster? **Index both** to 100 at a stated base
- How did B respond to A? A **connected scatter**

If a template forces one: columns for the quantity, a line for the rate, both axes from zero, no horizontal gridlines, each axis title colored to its series.

### The same data, story one

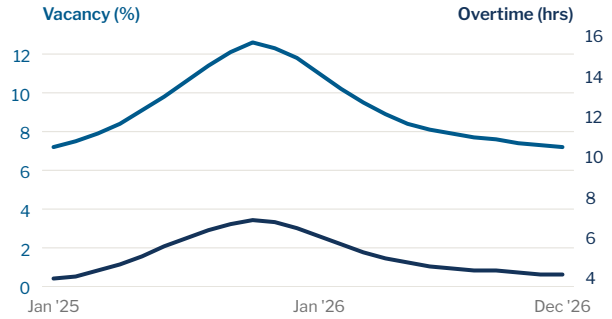
Ranges picked so the lines lie on top of each other: "overtime tracks vacancies exactly".




**Try to avoid**
The fit is an artifact of two chosen ranges.

### The same data, story two

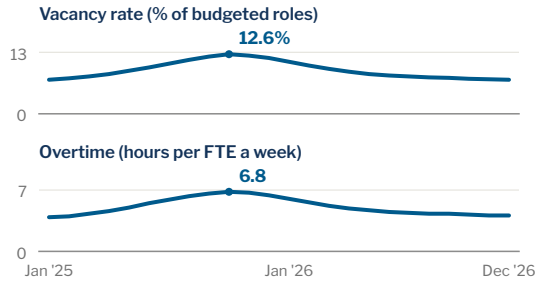
Change the right-hand range and overtime now looks flat while vacancies soar.




**Try to avoid**
Two scales, two contradictory stories.

### Two aligned panels

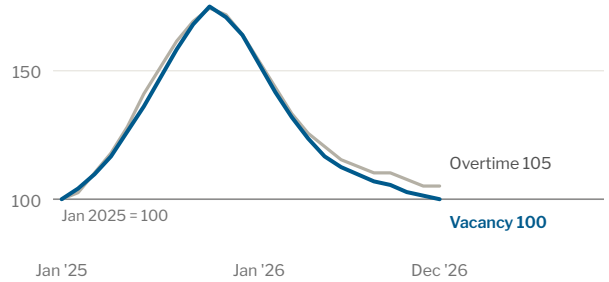
Each measure on its own scale from zero; the shared time axis makes timing exact. Both peaked in Oct 2025.




**Recommended**
When the question is timing.

### Indexed to 100

One honest axis: by Oct 2025, vacancies were up 75% and overtime 74%.




**Recommended**
When the question is relative change.

CORRELATION

# **Diverging** heatmap

A heatmap shows one measure across two sets of categories as color, so the eye finds patterns in 30 to 150 cells at once: a weak column, a strong row, one hot cell.

Use a diverging scale when the data has a meaningful middle: a benchmark, a target, change vs. last year. Use a sequential scale when the reference sits at one end, such as zero.

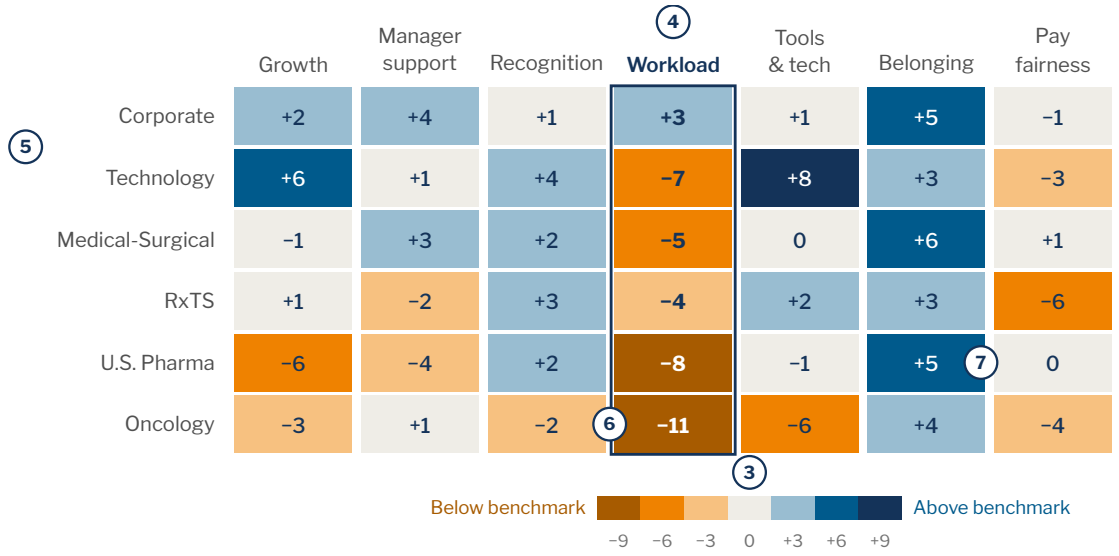
### McKesson spec

- Seven even classes; the middle one Light Sand
- Orange below the reference (worse), blue above
- Rows ranked by value, never alphabetical
- Outline one column or row; don't add a hue

1. **Insight headline**
   The pattern, counted from the data.
2. **Subtitle**
   Measure, reference, scope, unit.
3. **Neutral middle**
   Symmetric classes around zero, stepped key with end labels.
4. **Focus column**
   Outlined in navy, header bold.
5. **Ranked rows**
   By average gap, so the pattern emerges.
6. **Orange means worse**
   Below the benchmark, darker the further below.
7. **Values in the cells**
   White on dark fills, navy on light.
8. **Footnote and source**
   Define the gap; note any suppression.

## Workload trails the benchmark in five of six business units, Oncology most of all (1)

Engagement driver score vs. external healthcare benchmark, by business unit, 2026 (points) (2)




Gap = business-unit favorable score minus the external healthcare benchmark. Rows ranked by average gap; cells under 10 respondents would show grey, marked n<10.

Source: Employee engagement survey and external benchmark (synthetic data for illustration), 2026 (8)

CORRELATION

# Heatmap variants

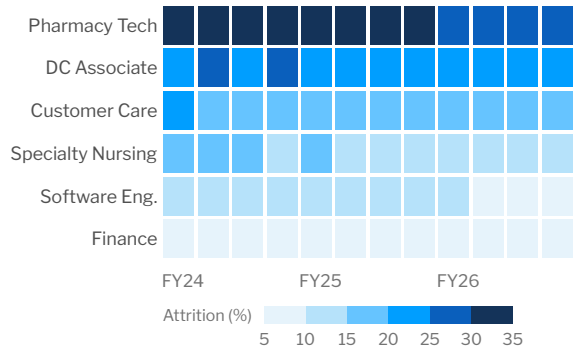
Same build: white gutters, even classes, ranked rows. Blue is the default ramp for any single measure.

### When a heatmap is the wrong chart

- A few values to compare precisely: a **bar** or **dot plot**
- A trend for one to four series: a **line**
- Columns in different units: a **table**, or one scale per column
- Places: a **map**

### Sequential, over time

Reference at one end: darkest = highest. Pharmacy Tech is darkest in every quarter.



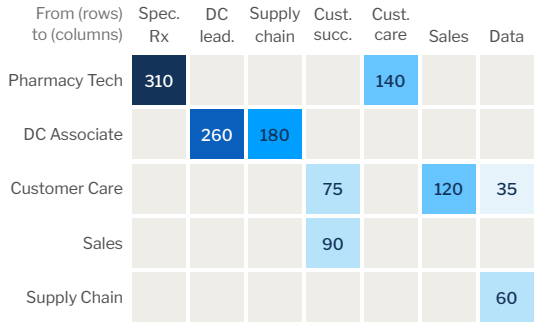

### Heat table

Values in every cell, color as a second cue. Oncology reached 11.2% in Q4 FY26.

| | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|
| **Oncology** | **8.9** | **9.6** | **10.4** | **11.2** |
| U.S. Pharma | 6.5 | 6.6 | 6.4 | 6.5 |
| Technology | 6.9 | 6.8 | 6.6 | 6.4 |
| Medical-Surgical | 6.3 | 6.1 | 6.4 | 6.2 |
| Corporate | 5.7 | 5.4 | 5.6 | 5.3 |
| RxTS | 4.9 | 5.1 | 4.8 | 4.9 |

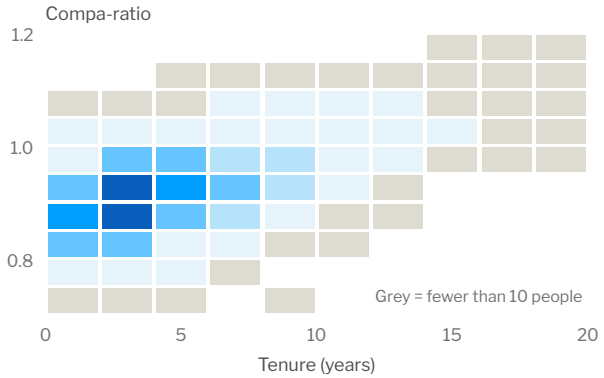
### Origin × destination

Many-to-many moves read better as a matrix than a tangled Sankey. Largest: Pharmacy Technician to Specialty Pharmacy Ops, 310 moves.




### Binned density

Thousands of people on two measures: count them into a grid instead of overplotting dots.



CORRELATION

# **Heatmap** best practice

Color is read less precisely than length, so the scale has to be honest and the order has to help.

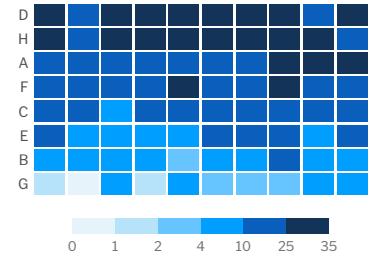
**Even steps.** Hand-picked breaks (1, 2, 4, 10, 25) make small differences look large. Use five to seven equal classes.

**Rank the rows.** Alphabetical order scatters the pattern; ranking by average reveals it.

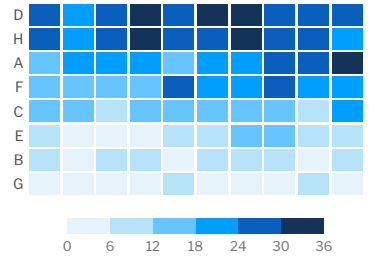
**Keep the zero step.** A diverging scale without a neutral class forces every cell above or below.

**No cell under 10 people.** Grey it, mark it n<10 and footnote it, in tooltips too.

## Color steps

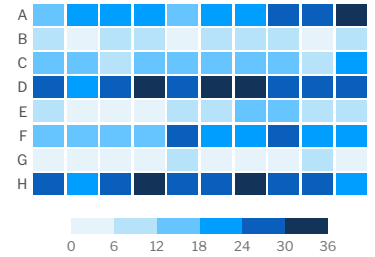



**Try to avoid**
Uneven breaks.

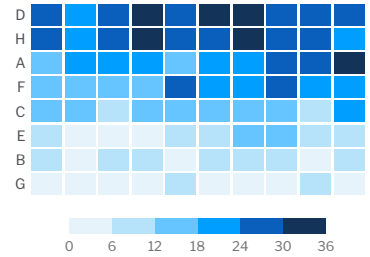



**Recommended**
Six equal classes.

## Sorting

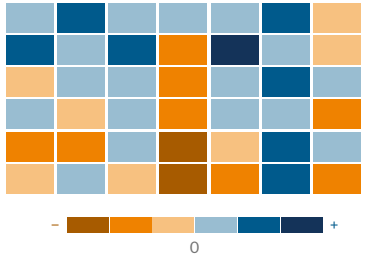



**Try to avoid**
Alphabetical rows.

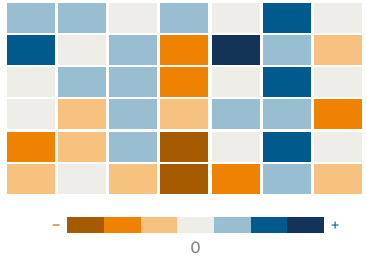



**Recommended**
Ranked by average.

## Zero on a diverging scale




**Try to avoid**
No neutral class.




**Recommended**
A Light Sand middle.

## Small groups

| | Growth | Mgr | Load | Tools |
|---|---|---|---|---|
| Inbound (46) | +4 | +2 | −3 | +1 |
| Outbound (38) | −2 | +1 | −6 | 0 |
| Returns (6) | −9 | −11 | −4 | −8 |
| Quality (21) | +3 | +5 | +1 | +2 |

**Try to avoid**
A six-person team scored.

| | Growth | Mgr | Load | Tools |
|---|---|---|---|---|
| Inbound (46) | +4 | +2 | −3 | +1 |
| Outbound (38) | −2 | +1 | −6 | 0 |
| Returns (6) | n<10 | n<10 | n<10 | n<10 |
| Quality (21) | +3 | +5 | +1 | +2 |

**Recommended**
Greyed and marked n<10.

CORRELATION

# **Build** correlation charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

scatter-bubble.md · dual-axis.md · heatmap.md

### Before you send it

- [ ] Bubble area ∝ value, with a size key
- [ ] Only the headline's points labelled
- [ ] Association, not cause, in the headline
- [ ] No dual axis: panels or an index
- [ ] Diverging midpoint = the real reference
- [ ] No point or cell under 10 people

| | |
|---|---|
| **PowerPoint and Excel** | **Bubble:** Insert › Insert Scatter (X, Y) or Bubble Chart › Bubble; split focus and context into two series; *Size represents: Area of bubbles*; label the focus series with *Value From Cells*. **Instead of a dual axis:** an index helper column `=B2/B$2*100`, or two line charts with identical width, aligned left. **Heatmap:** Conditional Formatting › Color Scales › More Rules › 3-Color Scale with fixed Number bounds (−10.5, 0, 10.5), or one rule per class; white cell borders as gutters. |
| **Power BI** | **Scatter chart** with the entity in Values; color by measure through *Markers* › *Colors* › *fx*; averages and thresholds as Analytics-pane constant lines. **Matrix** with *Cell elements* › *Background color* › *fx* (Gradient with a middle color, or Rules per class). Index measures replace a dual axis. Blank small groups: `IF([Headcount] < 10, BLANK(), …)`. |
| **Python** | matplotlib with `mckviz`; stepped classes from `BoundaryNorm`.<br>`ax.scatter(df.pay, df.exposure, s=df.headcount * K,  # area ∝ value`<br>`  c=mv.highlight(df.role, focus=focus), edgecolors="white")`<br>`norm = BoundaryNorm(np.arange(-10.5, 10.6, 3), 7)  # 7 even classes, 0 in the middle`<br>`ax.imshow(m, cmap=ListedColormap(mv.DIVERGING), norm=norm)` |
| **HTML** | Plotly `scatter` with `sizemode: "area"` and tooltips for the unlabelled points; `heatmap` with a stepped colorscale and `xgap`/`ygap` 2; `make_subplots(shared_xaxes=True)` for aligned panels. Avoid `secondary_y`. |
| **R** | `geom_point(aes(size = headcount))` + `scale_size_area()`; `geom_tile()` + `scale_fill_mck_div(limits = c(-10.5, 10.5))`; `facet_wrap(~ measure, ncol = 1, scales = "free_y")` for aligned panels. |

# Sankey diagram

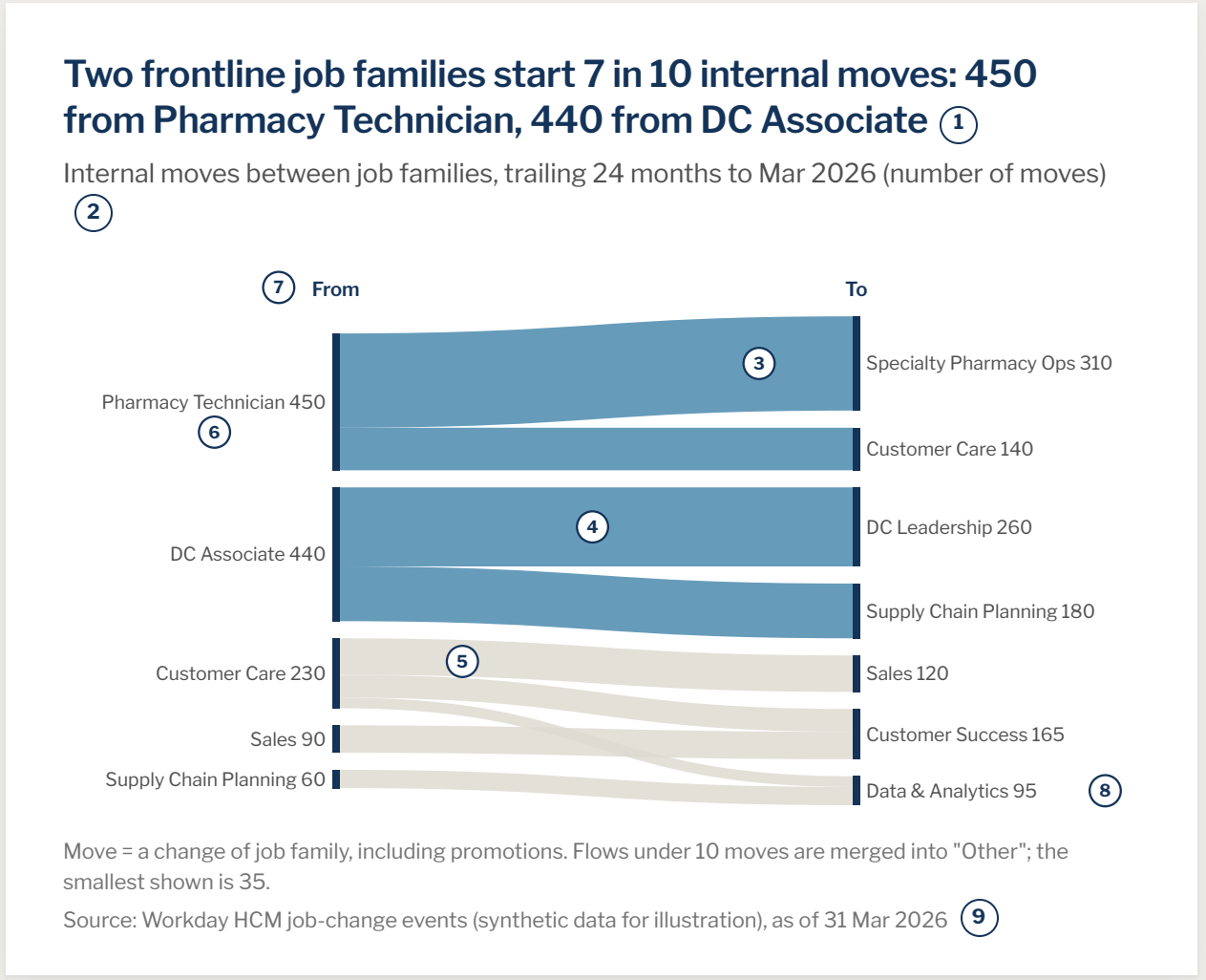
Relationship charts show how groups connect: how people move from one state to another, how they profile across many measures, or where groups overlap.

Each transfer is a band whose width is proportional to the people in it. Use it for internal mobility or a reskilling pipeline, when a few big flows matter most.

### McKesson spec

- Focus flows in blue; the rest light grey
- Thin navy node bars; name and total beside each
- Two to four stages, eight nodes a column at most
- No flow under 10 people: merge into "Other"

1. **Insight headline**
   Where the volume comes from, with the numbers.
2. **Subtitle**
   What flows, over what period, in what unit.
3. **Width = people**
   Every band on one shared scale.
4. **Focus flows in blue**
   The paths the headline names, drawn on top.
5. **Light grey context**
   Visible, but quiet.
6. **Nodes**
   Thin navy bars, name and total, no boxes.
7. **Stage headers**
   Bold, above each column.
8. **Ordered to untangle**
   Largest sources first; destinations follow their source.
9. **Footnote and source**
   Define a move; state the minimum flow.



# **Sankey** best practice

Color, shape and the n-size rule decide whether a Sankey can be read.

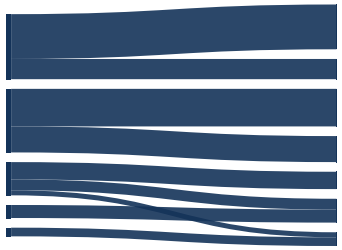
**Color the story.** One dark color makes crossing paths impossible to follow. Blue for the focus, light grey for the rest.

**Color transfer type, not stage.** The stages are already obvious. Give each type of transfer one color at every step; orange only for the loss.

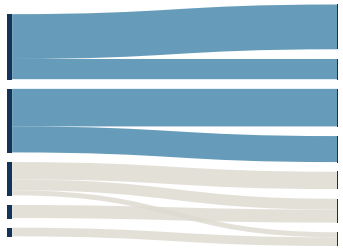
**Keep it organic.** Curved, semi-transparent bands and thin node bars. No boxed values, no angular arrows.

**No flow under 10 people.** A three-person move identifies people. Merge small flows into "Other".

## Coloring flows

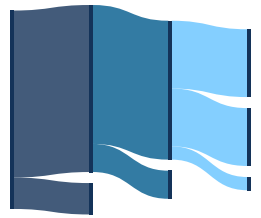


**Try to avoid**
One dark color.

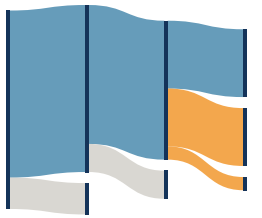


**Recommended**
The story in blue.

## Color by transfer type

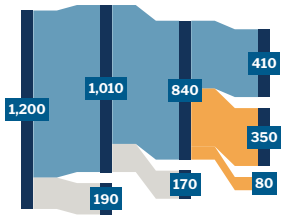


**Try to avoid**
A color per stage.

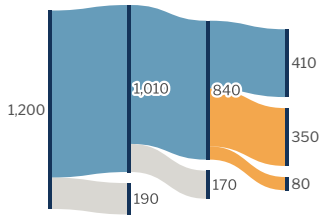


**Recommended**
Progressed, dropped, not placed.

## Styling

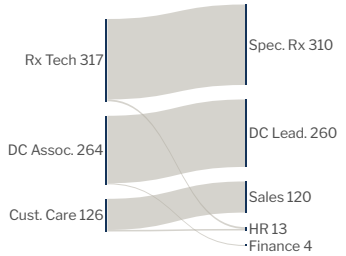



**Try to avoid**
Boxes and angles.

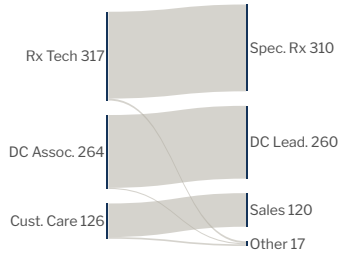



**Recommended**
Curves, plain numbers.

## Small flows




**Try to avoid**
Moves of 4, 6 and 7 people.




**Recommended**
Merged into "Other".

# **Parallel** coordinates

One vertical axis per measure, one line per group. It shows a profile: the shape a group makes across four to eight measures at once.

The axes are different measures, not points in time, so their order is a design choice. Put the story first, and orient every axis so up means the same thing.

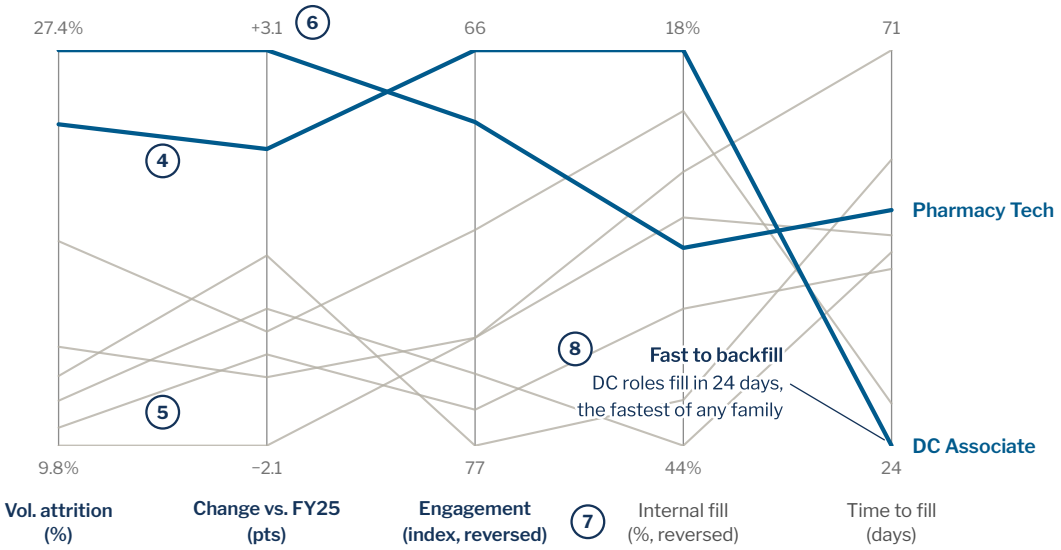
### McKesson spec

- One or two groups in blue; every other line grey
- Each axis scaled min–max, real min and max printed
- Reversed axes labelled; "up = worse" in the subtitle
- Every line a group of at least 10 people

1. **Insight headline**
   What the profile shows, measure by measure.
2. **Subtitle**
   Groups, measures, scaling and direction.
3. **Story axes first**
   Bold names; related measures side by side.
4. **Focus lines in blue**
   Heavier, on top, named at the right end.
5. **Grey context**
   The other families, thin and unnamed.
6. **Real range at the ends**
   Max on top, min below, with units.
7. **Reversed, and said so**
   So up always means worse.
8. **One callout**
   The exception worth a sentence.
9. **Footnote and source**
   Scaling, reversals, group sizes.

## Pharmacy Technicians and DC Associates are the two worst job families on attrition, its trend and engagement (1)

Profile of eight job families on five measures, FY26 (each axis scaled min–max; up = worse) (2)




Engagement and internal-fill axes are reversed so that up always means worse. Every line is a job family of 850+ people.

Source: Workday HCM, ATS and engagement survey (synthetic data for illustration), FY26 (9)

# Parallel coordinates best practice

The chart works only when one group stands out from a quiet background.

**Use it instead of a dense table.** A grid of 40 numbers hides the profile; lines show it at a glance.

**One group in color.** A shade of blue per line is a tangle with no message.

**Scale each axis.** On one raw scale, the measure with the biggest numbers flattens the rest.

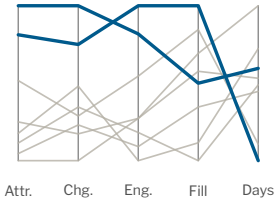
**Point every axis the same way.** Mixed directions make a consistently bad group zigzag.

Interactive versions can brush, reorder and hide axes; that beats cropping a static chart to two or three axes.

## Instead of a table

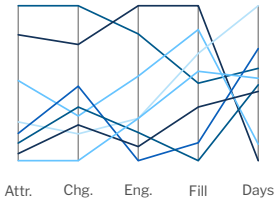
| | Attr. | Chg | Eng. | Int. | Days |
|---|---|---|---|---|---|
| Pharmacy | 27.4 | 3.1 | 68 | 31 | 52 |
| DC Associ | 24.1 | 1.8 | 66 | 18 | 24 |
| Customer | 18.9 | -0.6 | 71 | 22 | 29 |
| Specialty | 14.2 | -1.2 | 74 | 26 | 71 |
| Data & An | 12.9 | 0.4 | 77 | 41 | 58 |
| Supply Ch | 11.8 | -0.3 | 75 | 44 | 47 |
| Sales | 10.6 | -0.9 | 76 | 35 | 45 |
| Software | 9.8 | -2.1 | 74 | 29 | 49 |

**Try to avoid**
A grid of numbers.

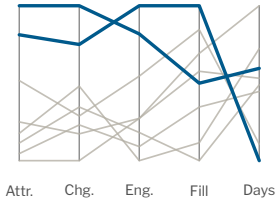



**Recommended**
The profile, highlighted.

## Highlighting lines

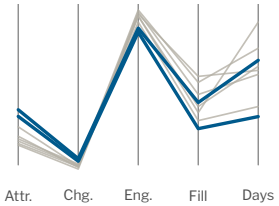



**Try to avoid**
A blue per line.

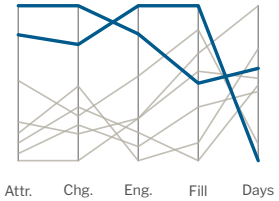



**Recommended**
Grey, one group blue.

## Scaling the axes

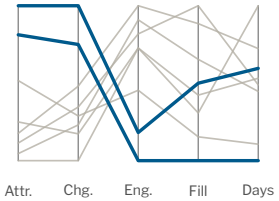



**Try to avoid**
One raw scale.

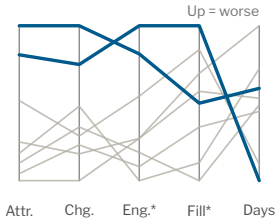



**Recommended**
Min–max per axis.

## Axis direction




**Try to avoid**
Up means better on some axes.




**Recommended**
Reversed axes marked *.

# **Venn** diagram

Overlapping circles show who sits in more than one group at once: people with several verified skills, mentors who also lead networks.

Use it for two or three sets when the overlap itself is the point. The circles don't encode size; the counts do. For four or more sets, or precise comparison, use combination bars.

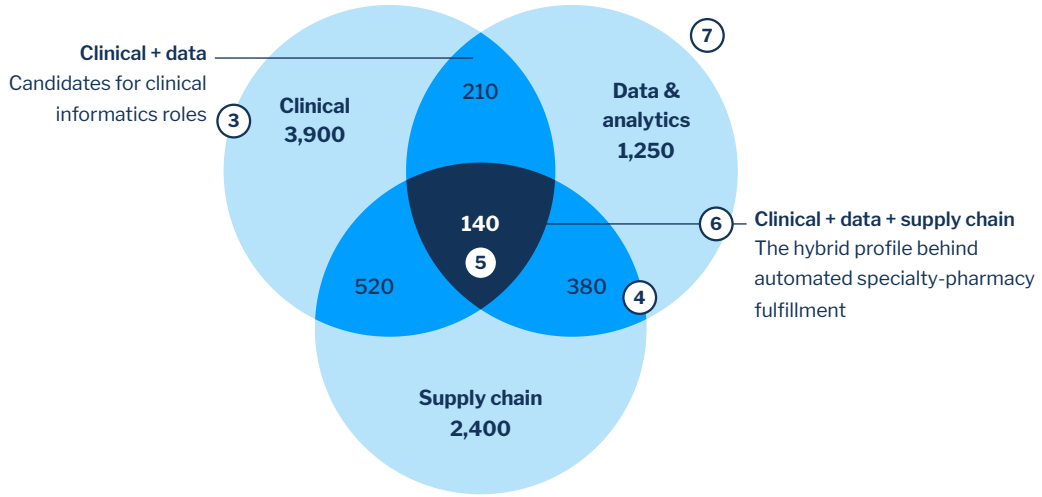
### McKesson spec

- One blue that darkens with each level of overlap
- Equal, regular circles; exact fills, no transparency
- Exclusive counts in each region; descriptions outside
- No region under 10 people

1. **Insight headline**
   The overlap that matters, against the total.
2. **Subtitle**
   What the sets are, who is counted, when.
3. **Single sets, lightest**
   Bold name, count beneath.
4. **Two-way overlaps**
   A darker blue; the count only.
5. **The center, darkest**
   Navy with white text.
6. **Descriptions outside**
   A bold title and a thin leader line.
7. **Equal circles**
   Not scaled to headcount; say so.
8. **Footnote and source**
   Define the sets; counts are exclusive.

## Only 140 of 8,800 employees with a verified skill profile combine clinical, data and supply-chain expertise (1)

Employees by verified skill area, U.S. regular employees, skills inventory as of Mar 2026 (headcount) (2)




Verified skill = manager-validated in the skills inventory. Circles are equal and not scaled to headcount; region counts are exclusive.

Source: Skills inventory (synthetic data for illustration), as of 31 Mar 2026; U.S. regular employees (8)

# **Venn** best practice

Keep the shapes simple and let the shading carry the logic.

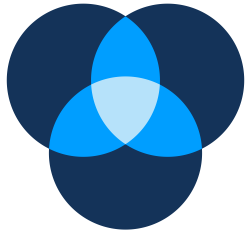
**Darker as sets overlap.** An overlap lighter than its sets reads backwards.

**Exact fills, one hue.** Translucent circles in different colors blend into muddy, off-brand tones.

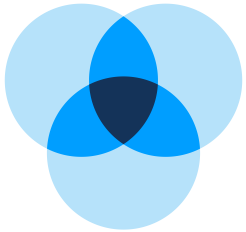
**Three sets at most.** Four ellipses make fifteen regions. Switch to combination bars.

**Text outside.** Sentences inside the circles hide the shading and can't be read.

## Shading overlaps

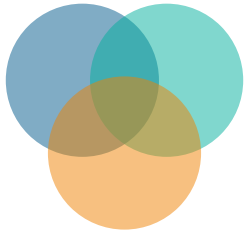


**Try to avoid**
Overlaps lighter.

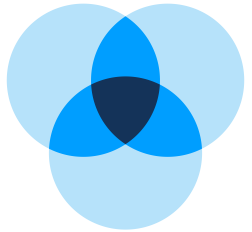


**Recommended**
Darker with each level.

## Fills



**Try to avoid**
Blended hues.

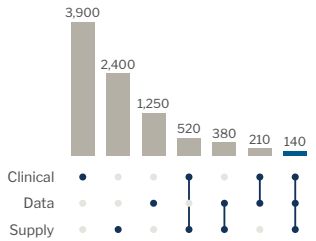


**Recommended**
Exact blues.

## Four or more sets

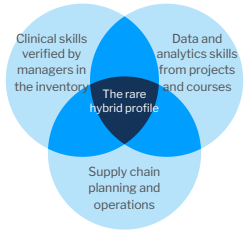


**Try to avoid**
Overlapping ellipses.

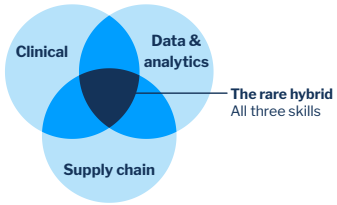



**Recommended**
Combination bars.

## Using text




**Try to avoid**
Sentences inside.




**Recommended**
Outside, with a leader.

# **Build** relationship charts

These charts are not native everywhere. Where a tool can't draw one, the cookbook says what to do instead.

sankey.md · parallel-coordinates.md · venn.md

### Before you send it

- [ ] Sankey flows conserved; exits are named nodes
- [ ] One color per transfer type; story in blue
- [ ] Axes scaled and oriented the same way
- [ ] Venn overlaps darker; counts exclusive
- [ ] No flow, line or region under 10 people

| | |
|---|---|
| **PowerPoint and Excel** | **Sankey:** no native chart. Build it in Python or HTML, export an SVG, then Insert › Pictures and *Convert to Shape* to edit labels; a linear pipeline is a native Funnel. **Parallel coordinates:** normalize in a helper table, Insert › 2-D Line, *Switch Row/Column* so measures are the categories, vertical gridlines as axes, focus series moved last. **Venn:** three Ovals; *Merge Shapes › Intersect* for each overlap; fills B6E2FA, 009EFF, 143359. |
| **Power BI** | No built-in Sankey: the AppSource Sankey visual only if your tenant has approved it; group small flows into "Other" in Power Query first. Parallel coordinates as a native Line chart with the measure on the X axis and a normalized DAX measure. No Venn visual: show the combinations as a sorted bar. |
| **Python** | Plotly `go.Sankey` for flows; matplotlib with `mckviz` for parallel coordinates and the Venn (polygon intersections, exact fills).<br>`link_color = {"progressed": mv.rgba(C.focus, 0.55),  # one color per transfer type`<br>`"dropped": mv.rgba(C.context, 0.45), "not_placed": mv.rgba(C.negative, 0.65)}`<br>`go.Sankey(node=dict(color=C.total, thickness=6),`<br>`link=dict(source=src, target=dst, value=people, color=[link_color[t] for t in kind]))` |
| **HTML** | Plotly `Sankey` with semi-transparent links; `Parcoords` lets readers brush and reorder axes; the Venn as filled `path` shapes with hover off. |
| **R** | `ggalluvial::geom_alluvium()` for Sankey-style flows; `geom_line()` over min–max scaled long data for parallel coordinates; `sf::st_intersection()` for Venn regions. |